Computer Science and Mathematics
Project Module Code
COC255

Semi-supervised medical image segmentation model using Consistency Regularisation

by

Sean C. Biscoe
F325508

Supervisor: Dr. Q. Meng

Department of Computer Science
Loughborough University

Winter 2025

Abstract

Lung cancer is one of the deadliest forms of cancer, which makes finding nodules in 3D CT scans as early as possible vital. While computer models can help, they usually need thousands of hand-drawn masks to learn properly, and getting these from doctors is difficult and time-consuming. This project looks at a way to train a high-quality model using far fewer labels by making use of ”unlabelled” data.

I built a 3D Residual U-Net and trained it using a Mean Teacher setup. This allowed the model to learn from a small group of 300 scans with labels, while also picking up extra information from 135 scans that had no labels at all. To test if this actually worked, I compared it against a baseline model that only had access to the labelled data.

The results showed that the semi-supervised approach was much more effective. It reached a Dice score (overlap) of 81.77%, which was a clear jump from the baseline's 78.27%. Just as importantly, the Average Surface Distance—which measures how close the edges of the prediction are to the real nodule—improved from 1.55 to 1.09 voxels. This proves that we can get better results without needing more manual work from radiologists. It shows that semi-supervised learning is a practical solution for building medical tools when expert time is limited.

Contents

1 Introduction **5**
1.1 Problem Statement 5
1.2 Aims and Key Objectives 6
1.2.1 Project Outcomes 6

2 Work plan with Gantt Chart **7**
2.1 Original Gantt Chart 7
2.2 Actual Gantt Chart 8

3 Literature review **9**
3.1 Consistency Regularisation 9
3.1.1 U-net 10
3.1.2 Mean Teacher 14
3.2 Semi-supervised medical image segmentation 16
3.3 Lung Cancer 17
3.3.1 Datasets 17
3.3.2 Methods using pseudo-labelling 18
3.3.3 Methods using Consistency-Regularisation 18
3.4 CT sign classification Models 19

4 Methodology **25**
4.1 Data description 25
4.2 Preprocessing 26
4.3 Dataset and Batching 27
4.3.1 3D Patching 27
4.3.2 Sampling Strategy 27
4.3.3 Tensor formatting and patch preparation 28
4.4 Model Architecture 28
4.4.1 Baseline U-Net architecture 28
4.4.2 Mean Teacher semi-supervised framework 31
4.4.3 Consistency loss design 33
4.4.4 Training setup and hyperparameters 34
4.4.5 Comparison of training setups 35
4.4.6 Parameter Opitmization 35

5 Experiments and Results **36**
5.1 Experimental Design 36
5.2 Evaluation Metrics 37
5.3 Baseline Performance 37

5.4 Semi-supervised Model Performance 38
5.5 Qualitative Segmentation Examples 39

6 Discussion **43**
6.1 Benefits and Limitations of the Approach 43
6.1.1 Benefits 43
6.1.2 Limitations 43
6.2 Further Interpretation of scans 44
6.3 Comparison with Related Work 44
6.4 Implications for Clinical Practice 45

7 Conclusion and Future Work **47**
7.1 Summary of Contributions 47
7.2 Overall remarks 47
7.3 Future Improvements 48

List of Figures

1 Architecture of the U-Net structure. 21
2 Overlap-tile strategy for large image segmentation. 21
3 Segmentation and weight map visualization for HeLa cell data. 21
4 Segmentation results on ISBI cell tracking challenge datasets. 22
5 Effect of noise and consistency on binary classification. 22
6 Diagram of the Mean Teacher training framework. 23
7 Comparison of classification cost and error on SVHN. 23
8 Hyperparameter sensitivity analysis on the SVHN dataset. 24
9 CT characteristics and radiological signs of pulmonary nodules. 24

10 Diagram of the Supervised Baseline training pipeline. 28
11 Diagram of the Mean Teacher semi-supervised training pipeline. 32

12 Comparison of Validation Dice Coefficient over 100 epochs. 38
13 Quantitative comparison of Baseline and Mean Teacher results across four metrics. 39
14 Dice distribution comparison for Baseline and Mean Teacher models. . . 40
15 Comparison of per-scan Dice scores (Baseline vs Mean Teacher). 41
16 Qualitative segmentation comparison of Baseline and Mean Teacher models on scan 730. 42

List of Tables

1 Ranking on the EM segmentation challenge Cell Tracking Challenge [2014] (march 6th, 2015), sorted by warping error. 12

2 These are the segmentation results (IOU) on the ISBI cell tracking challenge 2015. 13

3 This consists of the error percentage over 10 runs on SVHN with extra unlabelled training data 16

4 Performance comparison of the MLC models trained from scratch 20

5 Comparing the two frameworks set up 35

6 Quantitative comparison of Supervised Baseline and Mean Teacher on the LUNA16 validation set 37

7 Quantitative comparison of nodule segmentation performance on LUNA16. 45

Chapter 1

Introduction

1.1 Problem Statement

Lung cancer remains one of the most significant global health challenges. According to Zhan, Wang, Zhuo, Zhan, Yan, Shan, Zhou, Chen, and Liu [2023], lung cancer is the second most commonly diagnosed cancer worldwide and the leading cause of cancer-related mortality from Chhikara and Parang [2023]. Despite advances in treatment, the overall five-year survival rate remains low at approximately 22% Siegel, Miller, Fuchs, and Jemal [2022], largely due to late-stage diagnosis. As a result, early detection and accurate assessment of lung nodules are critical for improving patient outcomes.

Computed tomography (CT) imaging plays a central role in the early screening of lung cancer, enabling radiologists to assess pulmonary nodules based on characteristics such as size, shape, and texture. While CT screening has significantly improved early detection rates, the growing volume of imaging data places substantial demands on radiologists, increasing workload and the potential for diagnostic variability.

Automated computer-aided diagnosis (CAD) systems offer a promising solution to these challenges by assisting clinicians in the detection and classification of lung nodules. However, the development of robust deep learning models is often hindered by the limited availability of high-quality annotated medical data. To address this limitation, semi-supervised learning (SSL) techniques leverage both labelled and unlabelled data to improve model generalisation.

Furthermore, Deep learning models for lung nodule segmentation often struggle due to the lack of high-quality annotated data. Manually labelling 3D CT volumes is time-consuming, requires specialist radiologists, and varies between experts Setio et al. [2017]. This makes semi-supervised learning a natural fit as it can use the vast amounts of unlabelled CT scans from routine lung cancer screening to boost a small set of labelled data.

In this work, a 3D Residual U-Net is implemented within a semi-supervised Mean Teacher framework to address the volumetric segmentation of lung nodules in CT scans. By combining labelled and unlabelled scans through consistency regularisation, the proposed approach aims to improve diagnostic accuracy and provide more detailed clinical insights while reducing the annotation workload on specialist radiologists.

1.2 Aims and Key Objectives

To develop and evaluate a semi-supervised medical image segmentation model using Consistency Regularisation, demonstrating how unlabelled data can improve model performance when labelled data is limited.

Objectives:

1. Implement a supervised baseline using the U-Net architecture trained on a small, labelled subset of data.
2. Develop a Consistency Regularisation-based model, implementing the Mean Teacher framework.
3. Train and optimize the CR-based model using both labelled and unlabelled medical images.
4. Evaluate and compare performance between the supervised baseline and the CR model using standard segmentation metrics.
5. Analyse the impact of unlabelled data, consistency weight, and perturbation strength on model accuracy and stability.

1.2.1 Project Outcomes

This project will produce a working Mean Teacher model for semi-supervised segmentation using consistency regularisation. It will show how this beats a standard supervised U-Net when you don't have much labelled data. The work will give some useful pointers on how unlabelled data and consistency weighting help the model generalise better. Finally, it will leave behind a complete pipeline, which includes: preprocessing, training code, evaluation.

Chapter 2

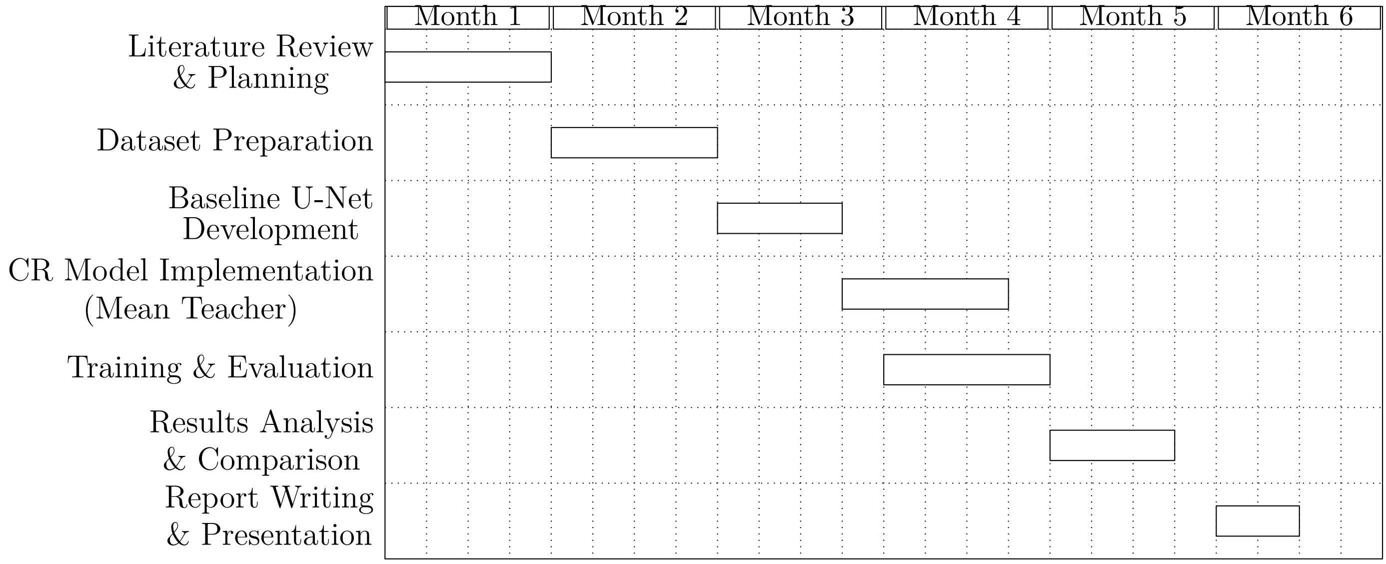
Work plan with Gantt Chart

2.1 Original Gantt Chart

In order to make sure I am able to stay on track I have created a list of tasks which I would like to complete in order in the table below.

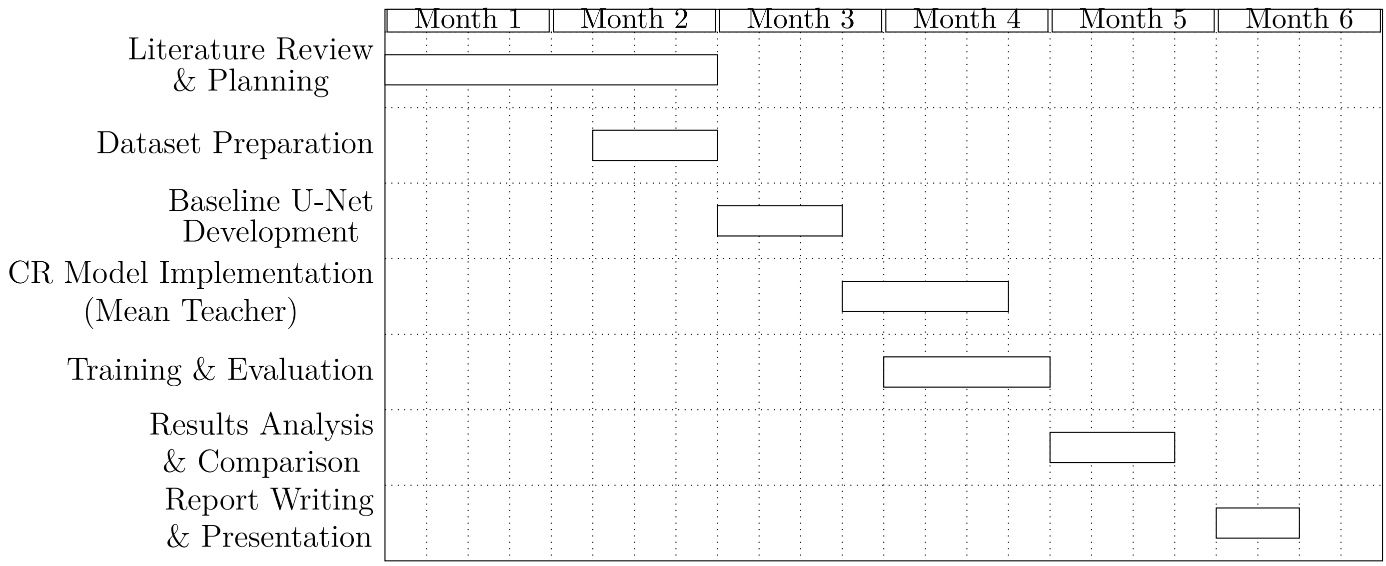
Phase	Tasks	Duration (Weeks)	Timeline (Months)
1. Literature Review & Planning	Review semi-supervised segmentation papers (U-Net, Mean Teacher, Lung Cancer), define objectives, install environment	4	Month 1
2. Dataset Preparation	Acquire datasets (LUNA16), preprocess (.nii → .h5), normalize, crop, and augment data	4	Month 2
3. Baseline Model Development	Implement and train a supervised U-Net baseline with limited labelled data	3	Month 3
4. Consistency Regularisation Model (Mean Teacher)	Implement CR framework with student teacher networks, EMA updates, and consistency loss	4	Month 3-4
5. Model Training & Evaluation	Train CR model, fine-tune hyperparameters, evaluate on validation/test sets	4	Month 4-5
6. Results Analysis & Comparison	Compare performance (Dice, Hausdorff metrics), visualize segmentation outputs, analyse effect of unlabelled data	3	Month 5
7. Report Writing & Presentation	Write final dissertation, prepare figures, presentation slides, and documentation	2	Month 6

I have put these tasks in the gaunt chart below so it becomes easier to visualise.



2.2 Actual Gantt Chart

Below is the Gantt chart with the actual time I have taken for each task and the time frame I started and completed each task.



Chapter 3

Literature review

3.1 Consistency Regularisation

Consistency Regularisation is the technique that encourages a model to produce stable predictions when its inputs or internal representations are perturbed Miyato, Kataoka, Koyama, and Yoshida [2018], which in practice means the process of randomly adding noise to the input data or model representations Li et al. [2021b]. This means that the predictions generated by the models under the same input data will be similar. Consistency Regularisation has become an important technique in semi-supervised learning, and a number of novel methods have been proposed based on this principle.

For example, Luo, Chen, Song, and Wang [2021] calculated the dual task consistency between the predictions of the pixel-level classification layer and the layer of level-set regression of the neural network. Li, Yu, Chen, Fu, Xing, and Heng [2021c] performed transformations such as translation, rotation, and flipping on the input images and calculated the transformation loss between the predictions they made. Duan, He, Zhang, Wang, Li, and Nie [2024] proposed Prediction Consistency Regularisation (PCR) for generalized category discovery, enforcing compatible predictions between a clustering-based classifier and a linear classifier in both label and feature space. One of the main inspirations for my project was Chen, Zhang, Mohiaddin, Wong, Firmin, Keegan, and Yang [2022], who proposed an adaptive hierarchical duel consistency framework based on the cross-domain data provided for the Left Atrium segmentation, to minimise effective information between and within domains.

Li, Zhao, Xu, Zeng, and Guan [2021a] proposed hierarchical consistency Regularisation based on the Mean-Teacher framework, which allows them to perform multi-level depth monitoring on the hidden layers of the neural network, so they are able to access finer information. The study by Wang, Peng, Pedersoli, Zhou, Zhang, and Desrosiers [2021a] was particularly notable as it not only proposed a self-paced and self-consistency method for semi-supervised segmentation, for the first time it also introduced time-aware consistency in co-training.

The study's above focus on consistency for semi supervised learning and also mentioned Mean-Teacher algorithm, which related to U-net. Shak, Al-Shabi, Liew, Lan, Chan, Ng, and Tan [2020] has showed how recently there has been advanced methods in semi-supervised field typically using the idea of Consistency Regularisation. This is based on the manifold assumption, which states that perturbations applied to the input does not cause a change in model output Yang, Song, King, and Xu [2023]. We use the method of consistency loss term when incorporating the idea of consistency Regularisation into semi-supervised methods. More specifically, for an unlabelled sample x_u, we define the consistency loss as follows:

$$\left\| P\big(y \mid \alpha(x_u), \theta\big) - P\big(y \mid \alpha'(x_u), \theta\big) \right\|_2^2$$

where $P(y \mid x, \theta)$ is used to donate the output distribution of the model whose parameter is θ, and we use α and α' to denote the random perturbation applied to the sample. The inputs of the model are not the same due to random perturbation which means there will be a difference in the outputs. We use the L2 loss to measure the difference (i.e. the consistency loss), and the Kullback-Leiber (KL) divergence loss as shown in study Bachman, Alsharif, and Precup [2014] and also cross-entropy loss mentioned in Xie et al. [2020a] and ReMixMatch Berthelot et al. [2019].

3.1.1 U-net

U-net is a Convolutional neural network (CNN) developed for image segmentation Ronneberger, Fischer, and Brox [2015]. This is used to identify the localisation, which is a class label that is supposed to be assigned to each pixel. However, in the past, this would require thousands of training images, which are unfortunately normally beyond reach in biomedical tasks, and so would also require too much processing resources, including time. This is why earlier approaches like Ciresan, Gambardella, Giusti, and Schmidhuber [2012] had a network that he trained in a sliding-window setup to predict the class label of each pixel and did this by providing a local region(patch) around that pixel as input. This means that this network can be localised. Then in terms of patches the training data is much larger than the number of training images. However, there are 2 main drawbacks with the strategy. The first is that the Convolutional network will be quite slow due to the network being run separately for each batch, and the overlapping patches cause a lot of redundancy. The second is that there is a trade off between the use of context and localisation accuracy. This is due to larger patches requiring more maximum pooling layers, which reduce the localisation accuracy, whilst the small patches allow the network to only see small context.

However, in Ronneberger, Fischer, and Brox [2015], the authors tried to build a more elegant architecture which is the so-called ”fully Convolutional network” Long, Shelhamer, and Darrell [2015] except they tried to modify and extend the architecture such that it works with very few training images and yields more precise segmentations (Figure 1). The main idea in Long, Shelhamer, and Darrell [2015] is to help a normal contracting network by successive layers, where the pooling operators are replaced by upsampling operators. So, these layers increase the resolution of the output. We need high resolution features from the contracting rather that are combined with the upsampled output in order to localise. This means that a successive convolution layer has the ability to learn to assemble a more precise output based on the information.

In the Ronneberger, Fischer, and Brox [2015] paper they have changed the architecture so that in the upsampling part they also have a large number of feature channels, which means that the network is allowed to propagate context information to higher resolution layers. This means that the expansive path is more or less symmetric to the contracting path and yields a u-shaped architecture. The network only uses the valid part of each convolution, i.e. the segmentation map only contains pixels so that the full context is available in the input image, and the network also does not have fully connecting layers. This strategy allows for the seamless segmentation of arbitrarily large images using an overlap-tile strategy (Figure 2). This means that to predict the pixels in the border region of the image, we mirror the input image to extrapolate the missing context. So, the tiling strategy is imperative to apply the network to large images as otherwise the GPU memory would limit the resolution.

When we have very little training data available, we need to use excessive data augmentation such as elastic deformations (as used in the original U-Net) to the available training images. We do this so that the network is allowed to learn invariance to such deformations, without the need to see these transformations in the annotated image corpus. This is particularly important in biomedical segmentation and is so very important for my project, since we can use the deformation to be the most common variation in tissue, and realistic deformations can be simulation effective. In Dosovitskiy, Springenberg, Riedmiller, and Brox [2014] we can see the value of data augmentation for Learning invariance in the scope of unsupervised feature learning.

They proposed the use of weighted loss, in which you separate the background labels between touching cells so that you can obtain a larger weight in the loss function. They did this as a solution to the challenge of separation of touching objects in the same class Figure (3). The network architecture of as illustrated in Figure 1 consists of a Contracting path (left side) and an expansive path (right side). The left side follows the typical convolutional network architecture which includes the repeated application of the two 3x3 covulations (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a2x2 max pooling operation with stride 2 for downsampling. We double the number of feature channels at each downsampling step. In the expansive path, every step includes an upsampling of the feature map followed by a 2x2 convolution ("up-convolution") that half the number of feature channels, which is a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. At the final layer a 2x2 convolution is used to map each 64-component feature vector to the desired number of classes. The network has 23 convolutional layers in total. We need to select the input tile size such that all 2x2 max-pooling operations are applied to a layer with an even x- and y-size to ensure that seamless tiling of the output segmentation map occurs ((Figure 2)).

We can train the network with input images and their corresponding segmentation maps with stochastic gradient descent implementation of the paper used by Jia, Shelhamer, Donahue, Karayev, Long, Girshick, Guadarrama, and Darrell [2014]. Ronneberger et al. minimized the overhead and made maximum use of the GPU memory by favouring large input files over a large batch size so the authors ended up reducing the batch size to a single image. The energy function is computed by a pixel-wise soft-max over the final feature map combined with the cross entropy loss function. We define the soft-max as

$$p_k(\mathbf{x}) = \frac{\exp\big(a_k(\mathbf{x})\big)}{\sum_{k'=1}^{K} \exp\big(a_{k'}(\mathbf{x})\big)}$$

Table 1: Ranking on the EM segmentation challenge Cell Tracking Challenge [2014] (march 6th, 2015), sorted by warping error.

Rank	Group name	Warping Error	Rand Error	Pixel Error
	** human values **	0.000005	0.0021	0.0010
1.	u-net	**0.000353**	0.0382	0.0611
2.	DIVE-SCI	0.000355	0.0305	0.0584
3.	IDSIA [2]	0.000420	0.0504	0.0613
4.	DIVE	0.000430	0.0545	**0.0582**
⋮				
10.	IDSIA-SCI	0.000653	**0.0189**	0.1027

where $a_k(\mathbf{x})$ denotes the activation in the feature channel k at the pixel position $\mathbf{x} \in \Omega$ with $\Omega \subset \mathbb{Z}^2$. K is the number of classes and $p_k(\mathbf{x})$ is the approximated maximum-function. So $p_k(\mathbf{x}) \approx 1$for k that has the maximum activation $a_k(\mathbf{x})$ and $p_k(\mathbf{x}) \approx 0$ for all other k. Then the cross entropy will penalize at each position the deviation of $p_{\ell(\mathbf{x})}(\mathbf{x})$ from 1 using

$$E = \sum_{\mathbf{x}\in\Omega} w(\mathbf{x}) \log\big(p_{\ell(\mathbf{x})}(\mathbf{x})\big)$$

where $\ell : \Omega \to \{1, \dots, K\}$ is the true label of each pixel and $w : \Omega \to \mathbb{R}$ is w weight map introduced to give some pixels more importance in the training. We can then force the network to learn the small separation borders that we introduce between touching cells (see Figure 3c and 3d) using morphological operations. We pre-compute the weight map for each ground truth segmentation to compensate the different frequency of pixels from a certain class in the training data set. The weight map is computed as

$$w(\mathbf{x}) = w_c(\mathbf{x}) + w_0 \cdot \exp\left(-\frac{\big(d_1(\mathbf{x}) + d_2(\mathbf{x})\big)^2}{2\sigma^2}\right)$$

where we: $w_c : \Omega \to \mathbb{R}$ is the weight map to balance the class frequencies, $d_1 : \Omega \to \mathbb{R}$ denotes the distance to the border of the nearest cell and $d_2 : \Omega \to \mathbb{R}$ the distance to the border of the second nearest cell.

The table above, 1, shows how during experiments the u-net (arranged over 7 rotated versions of the input data) achieves without any further pre- or post processing a warping error of 0.0003529, which is the new best score, and a rand-error of 0.0382. This evaluation was done by thresholding the map at 10 different levels and computation of the "warping error", the "Rand error" and the pixel error from experiment done by Arganda-Carreras, Turaga, Berger, Cireşan, Giusti, Gambardella, Schmidhuber, Laptev, Dwivedi, Buhmann, Liu, Seyedhosseini, Tasdizen, Kamentsky, Burget, Uher, Tan, Sun, Pham, Bas, Uzunbas, Cardona, Schindelin, and Seung [2015]. The U-net errors are significantly better then the sliding-window convolution network result by Ciresan, Gambardella, Giusti, and Schmidhuber [2012], whose best submission had a warping error of 0.000420 and a rand error of 0.0504.

In the next table, 2, we can see that u-net achieved an average IOU(”intersection over union”) of 92%, which is significantly better than the second best algorithm with 83% with the first data set ”PhC-U373”. This was achieved when they applied the u-net algorithm to a cell segmentation algorithm task in light microscopic images, in the segmentation task apart of the ISBI cell tracking challenge 2014 and 2015 Maška et al. [2014], Cell Tracking Challenge [2014]. The first data set ”PhC-U373” contained 35 partially trained annotated training images of Gliobastoma-astrocytoma U373 cells on a polyacrylimide substrate recorded by phase contrast microscopy (see Figure 4a,b). The second data set u-net average IOU achieved was 77.5% significantly better than the second best algorithm with 46%. This data set is ”DIC-HeLa” which contains HeLa cells on a flat glass recorded by differential interference contrast (DIC) microscopy (see Figure 3, Figure 4c,d) and 20 partially annotated images.

U-net performed extremely well across different biomedical segmentation tasks and outperformed several sliding-window CNNs and achieving state-of-the-art results on EM and ISBI challenges. So using a type of U-net algorithm such as the Mean teacher Method will be suitable for my Project. This is due to with the help from data Augmentation, which I have explained below allowing u-net to only need a few annotated images to perform well.

Data Augmentation

This is the technique used to teach the network the desired invariance and robustness properties, when only few training samples are available. When using microscopic images we primarily need shift and rotation invariance as well as robustness to deformations and gray value variations. To train a segmentation network with very few annotated images using random elastic deformations of the training samples seems to be key. Then we generate smooth deformations using random displacement vectors on a coarse 3 by 3 grid.

3D Extension for LUNA16

For volumetric CT data like LUNA16, the 2D U-Net is extended to 3D with $3 \times 3 \times 3$ convolutions and $2 \times 2 \times 2$ pooling, maintaining the encoder-decoder structure with skip

Table 2: These are the segmentation results (IOU) on the ISBI cell tracking challenge 2015.

Name	PhC-U373	DIC-HeLa
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	-
second-best 2015	0.83	0.46
u-net (2015)	**0.9203**	**0.7756**

connections. This preserves the few-shot learning capability while handling the increased memory demands of 3D medical volumes.

3.1.2 Mean Teacher

Tarvainen and Valpola [2018] uses the fact that when they change a percept slightly, the human will still consider it to be the same object. So, a classification model should favour functions that give a consistent output for similar data points. An approach for this is to add noise to the input of the model. We can make the model learn more abstract invariances by adding noise to intermediate representations, which is what has motivated many Regularisation techniques, for example Dropout in the study by Srivastava et al. [2014]. We could minimize the classification cost at the zero-dimensional data points of the input space, but we can push decision boundaries away from the labelled data points by the regularized model pushing away costs; see figure 5b.

The noise Regularisation by itself does not aid in semi-supervised learning due to the classification cost not being defined for unlabelled examples. We can use the R model in the study done by Rasmus et al. [2015] that evaluates each data point with and without noise, and then applies a consistency cost between the two predictions to move past this. Here we will have the model assuming a dual role as a teacher and a student. As a teacher, it will generate targets that are then used by itself as a student to learn. As the model generates targets, they may be incorrect. Also, if there is too much weight given to the generated targets, the cost of inconsistency will outweigh the cost of misclassification, which in turn prevents the learning of new information. This means that the model will suffer from confirmation bias; see Figure 4c, when we improve the targets, a hazard can be mitigated.

We can improve the quality of the target in at least two ways. The first way is that, instead of barely applying additive or multiplicative noise, we can choose the perturbation of the representations. The second approach is to choose the teacher model carefully instead of barely replicating the student model. In the study by Miyato et al. [2017] they took the first approach and showed that Virtual Adversarial Training can yield impressive results. This is why, Tarvainen and Valpola [2018] took the second approach to show that it also provides significant benefits. We can potentially combine these two approaches to produce even better outcomes as well.

In the study Tarvainen and Valpola [2018], the goal is to form a better teacher model from the student model without additional training. They will consider that the softmax output of a model does not usually provide accurate predictions outside of the training data. You can partially alleviate this by adding noise to the model such as inference timing in Gal and Ghahramani [2016], and then a noisy teacher theoretically can produce more accurate targets (Figure 4d). We have seen this approach being used in Bachman, Alsharif, and Precup [2014], and also in semi-supervised image classification Laine and Aila [2017] and Sajjadi, Javanmardi, and Tasdizen [2016]. Laine and Aila [2017] mentioned the Pie model that Tarvainen and Valpola [2018] used as the basis for their experiment. The Pie Model can be further improved by Temporal Ensembling in Laine and Aila [2017].

The Mean Teacher method is based off the Temporal Ensembling method except that we use averaging model weights not predictions. Temporal Ensembling maintains an exponential moving average (EMA) of label predictions on each training example and penalizes if predictions are inconsistent with this target. The targets will only change once per epoch, meaning Temporal Ensembling is very inefficient when learning larger data sets. This is why we use Mean Teacher as Mean Teacher improves test accuracy and allows for training with fewer labels than Temporal Ensembling.

The teacher is an average of consecutive student models, whilst using averaging model weights instead of predictions, so we can call this the Mean Teacher method (see Figure 6). When we average the model weights over the training steps instead of using the final weights directly, we tend to produce a more accurate model as found in the study performed by Polyak and Juditsky [1992]. So we are able to take advantage of this during training which allows us to construct better targets. The teacher uses the EMA weights of the student model instead of sharing the weights of the student. This allows the Model to aggregate information after every step instead of every epoch such as in Ensembling. As well as this the target has a better intermediate representations cue to the weight averages improving all layer outputs and not just the top output. This leads to two main practical advantages over Temporal Ensembling. First, there is better test accuracy due to more accurate target labels which lead to a faster feedback loop between the student and the teacher. Second, the approach scales to online learning and large datasets.

This means they defined the consistency cost J as the expected distance between the prediction of the teacher model (with weights θ' and noise η')) and the prediction of the student model (with weights θ and noise η))

$$J(\theta) = \mathbb{E}_{x,\eta',\eta}\Big[\big\| f(x,\theta',\eta') - f(x,\theta,\eta)\big\|_2^2\Big]$$

The difference between the models: Π , Temporal Ensembling and Mean Teacher method is how the teacher predictions are generated. The Π model uses $\theta' = \theta$, and Temporal Ensembling approximates $f(x,\theta',\eta')$ with a weighted average of successive predictions. Conversely, we define θ'_t at training step t as the EMA of successive θ weights:

$$\theta'_t = \alpha\theta'_{t-1} + (1-\alpha)\theta_t$$

where alpha is a smoothing coefficient hyperparameter. Another difference between the three algorithms is that the $\dddot{\Pi}$ model trains the θ' however the Temporal Ensembling and Mean Teacher treat θ' as a constant with regards to optimisation. To approximate the consistency cost function J we sample the noise, η, η' at each training step with stochastic gradient descent. They followed the study done by Laine and Aila [2017] and use the mean square error (MSE) as the consistency cost in most of their experiments.

From their research we have seen that the Mean Teacher improves the speed of learning and the classification accuracy of the trained network. As well as that, Mean Teacher scales very well compared to state-of-the-art architectures and large image sizes and is so better than Temporal Ensembling as works with large datasets and on-line learning. An example of this is that Mean Teacher uses unlabelled data more efficiently than the pie model. However the Pie model will keep improving for longer with 500K extra unlabelled samples. This is shown in the right hand column of Figure 7 but even then for the Mean

Teacher model the model learns faster and eventually converges to a more accurate result, but the vast amount of data would mean the the Pie models prediction will improve a lot. We can see this also in the results of Error percentage with the Mean Teacher Method compared to the Pie model in Table 3.

Limitations of the Mean Teacher model include: we require dropout or augmentation for effective regularisation. Also that input noise hardly benefits Mean Teacher if we have augmentation so it would be a waste, and that dropout on the teacher side provides only marginal benefit over just having it on the student side Tarvainen and Valpola [2018] as we can see in Figures 8a and 8b, for when input augmentation is in use.

Another limitation is decoupling and consistency, figure 8e. So when we decouple the Mean Teacher method and changed the model to have two top layers which produce two different outputs, and so we train one for consistency and the other for classification. The Strongly coupled result performed very well but the two loosely coupled versions do not. So I will ensure to use a very coupled Mean Teacher algorithm when I am working.

Extension to 3D LUNA16 Segmentation

For LUNA16 lung nodules, Mean Teacher uses identical 3D ResU-Net architectures for student and teacher. Consistency loss is computed as MSE between teacher and student predictions on unlabelled 64^3 patches, with supervised Dice loss on labelled patches. EMA updates ($\alpha = 0.999$) provide stable targets despite small-batch 3D training noise.

3.2 Semi-supervised medical image segmentation

From [Basak and Yin, 2023] we know that semi-supervised methods are being used more and more compared to Full-supervised methods, as they take less memory, less processing time, and require much less inputs. This means in the medical world, we do not require as many fully labelled medical image data sets. So, in the more recent years many semi-supervised methods have been applied to medical image segmentation; this is due to semi-supervised methods being able to use numerous unlabelled data to improve the performance of models through extensive training of the network when they have insufficient labelled data.

An example of this being Yu, Wang, Li, Fu, and Heng [2019b] improved the Mean-Teacher semi-supervised framework and introduced uncertainty-aware to evaluate the predictions for the Left Atrium Segmentation. In addition, Cui et al. [2019] has designed an adap-

Table 3: This consists of the error percentage over 10 runs on SVHN with extra unlabelled training data

	500 labels 73257 images	500 labels 173257 images	500 labels 573257 images
Π model (ours)	6.83 ± 0.66	4.49 ± 0.27	3.26 ± 0.14
Mean Teacher	**4.18 ± 0.27**	**3.02 ± 0.16**	**2.46 ± 0.06**

tive Mean-Teacher semi-supervised model for segmentation of Brain Lesion , where both student-teacher models were DeepMedic architecture. In the study Zhang and Zhang [2021], they proposed a dual-task learning framework that performed consistency Regularisation of the two task levels of the segmentation probability map and the signed distance map. Wang et al. [2021b] in his study promoted the interaction of segmentation and classification for semi-supervised Pancreatic Ductal Adenocarcinoma segmentation by applying induced attention guidance.

The training of a 3D model as mentioned in Shak et al. [2020] requires a large amount of labelled data and as I will mention in the MLC section of my Literature review, a nodule annotation is labour intensive, whilst the acquisition of unlabelled nodule data is competitively convenient. This is why the study used an SSL method to improve MLC performance. They were able to split the data into labelled and unlabelled data where the labelled data looked like:

$$X_l = \left\{ \left(x_l^{(i)}, y_l^{(i)} \right) \right\}_{i=1}^{N_l}$$

for a dataset with N_l samples where $x_l^{(i)}$ denotes the input sample, and

$$y_l^{(i)} = \left[y_1^{(i)}, \ldots, y_C^{(i)} \right] \subseteq \{0, 1\}^C$$

denotes the corresponding ground truth table, and C denotes the number of labels for MLC. $y_c^{(i)} = 1$ indicates the label exists in the i-th sample whilst $y_c^{(i)} = 0$ indicates the label does not exist in the i-th sample. Also $y_c^{(i)}$ denotes whether a particular sign is present in the sample. As well as this,

$$X_u = \left\{ \left(x_u^{(i)} \right) \right\}_{i=1}^{N_u}$$

denotes an unlabelled dataset with N samples. Therefore the entire dataset can be represented as $X = \{X_l, X_u\}$.
This is why I will be using an SSL method to reduce time having to label nodules and making my MLC performance improve.

3.3 Lung Cancer

3.3.1 Datasets

A big issue in the medical imaging field is finding labelled/annotated data that is sufficient to validate new deep learning methods Shak et al. [2020]. There are relatively small annotated datasets that include LUNA16 and LIDC-IDRI [Setio et al. [2017],Armato et al. [2011]] which are typically used for lung nodule detection or classification. The LUNA16 and LIDC-IDRI datasets contain detailed annotated locations of each lung nodule that was within the CT scans. From research the LIDC-IDRI dataset, there are potentially other signs that can affect the diagnosis of nodules, such as: spiculation, lobuation, vessel convergence (VC), and also pleural identification (PL) and air bronchogram (AB) often indicates a maligant nodule [Albert and Russell, 2009]. Unfortunately, the LIDC dataset will not be able to meet the needs of nodule signs.The National Lung Screening Tiral (NLST) lung screening dataset as shown in the study by Aberle et al. [2011], was released by the National Cancer Institute (NCI), is a large resource of lung CT scans;

however, an unfortunate downside of the NLST dataset is that the ground truth data will only produce a final diagnosis of the cancer (i.e. 1) or cancer-free (i.e 0) for each patient and will not provide the labelled/annotated locations of the nodules within the CT scans.

What this means for my project is due to the lack of availability of labelled datasets and even in those labelled datasets, there is not too much data to use, which naturally leads me to the use of SSL as then I do not require as many labelled datasets. This means that I will be using the LUNA16 dataset.

3.3.2 Methods using pseudo-labelling

To this date there are only a couple known methods which can simultaneously detect and classify nodes from computed Tomography (CT) scans as shown in Redmon, Divvala, Girshick, and Farhadi [2015]. However Shak et al. [2020] presents a new way which is end-to-end scheme to detect and classify lung nodules that is able to predict lung cancer, and evaluated on a comprehensive CT screening dataset of about 4,000 scans.

To overcome this insufficiency in the NLST dataset, we can use the Self-training with Noisy Student Training as shown in the study by Xie et al. [2020b] which allows leverage to the unlabelled data/nodes. In the study by in ImageNet Xie et al. [2020b], they found the most stat-of-the-art results on Image Classification was produced by Self-training. We can use the self-training method which uses a trained teacher model that will generate pseudo labels on unlabelled images. At this point a student model can be trained on both the labelled and pseudo-labelled images, and we can add noise simultaneously to improve the students learning capacity. So in the self-training network, they use state of the art Local-global network as shown in study Al-Shabi et al. [2019a] to classify unlabelled nodules in the NLST dataset and train a new student model with noise on the pseudo-labelled data. They will then obtain significant improvement in their lung nodule classification scheme in terms of the area beneath the receiver characteristic curve (AUC) and obtained state-of the art retulst for lung cancer prediction NLST dataset which is very simular to the results of the original self training paper Xie et al. [2020b].

Also there are many other deep leaning based detection models available including DeepLung, DeepSEED, NODULe, 3d-semi-supervised convolution transfer neural network and Lung Nodule detection using Faster R-CNN [Zhu et al. [2018],Li et al. [2020],Liu et al. [2019], Zhang et al. [2018]]. As well as this, deep learning modules have been shown to differentiate between maligant and benign nodules very well [Al-Shabi et al. [2019a], Al-Shabi et al. [2019b],Xie et al. [2019],Al-Shabi et al. [2022]].

3.3.3 Methods using Consistency-Regularisation

We can see in Figure 9 that there are certain signs we are able to recognise which are an important basis for radiologists to differentiate lung nodules in CT. This means that we can use accurate automatic classification of nodule signs to help doctors improve diagnostic effectiveness. There is also a certain degree of subjectivity in judging signs which means automatic classification can reduce errors caused by radiologists' subjective agreements. As well as this, end-to-end diagnostic modules are being increasingly applied within classification tasks as shown in studies by [Zhu et al. [2018],Al-Shabi et al. [2022]], however, it is not clear how these models work as shown by Wu et al. [2018].

CT sign classification models are divided into 3 types: binary classification models, multi classification models and multi label classification (MLC) models.

3.4 CT sign classification Models

Binary classification model

The binary classification model, is typically built for a certain sign. We can use hybrid resampling and feature fusion, we can design two classification models which are for then ground-glass opacity and the cavity in nodules as seen in studies [Han et al. [2018],Han et al. [2019]]. In another study, the snake model (active contour model) which was applied to extract the specific contour of a nodule and also to help build a spiculation recognition model as shown in Qiu et al. [2020]. For a certain CT sign the binary classification model will often make full use of the shape and texture characteristics and normally has good performance in the classification. Unfortunately, the model performance drops when the model is used for other signs. So if one model is designed for each sing, the overall classification effectiveness will be reduced.

Multiclassification model

The Multiclassification model is a model that only outputs one positive class from multiple classes. An example being in a study by Ma et al. [2015] a fused classification model is a model that can identify up to 9 kinds of lung CT signs. So we use image retrieval methods to realise CT sign classification [Ma et al. [2017],Ma et al. [2020]]. In study He [2019] he proposed that we use images of signs generated by a generative adversarial network (GAN) to pre-train the convolutional neural network (CNN) mode, so we can then use real data to fine-tune the model. We use classification sign datasets to train the multiclassification models on, so that each image in the dataset will only contain one sign. Unfortunately this leads to two issues: (I) the process will not align to the actual clinical application scenario and (II) the data labelling process will be extremely tedious. Unfortunately the mutlticlassification model is only able to output 1 positive class at a time and so cannot meet the needs.

Multilabel classification (MLC) mode

To be able to evaluate all the signs in the nodular lesion area simultaneously we need an MLC model as an MLC model can output multiple positive classes at once. This also would greatly reduce the annotation workload due to radiologists only requiring to annotate the nodules with global annotations of signs. This study proposed a new SSL training method of uncertainty-aware (UA) self-training framework with consistency Regularisation (USC). This method has been applied in the MLC task of nodule signs and can also be extended to multiple other simular scenarios.

They based their experiment on all the labelled data for the LMC of nodule signs and used the schemes of a 2D model and 3D models respectively. For the 2D model they used Local_Global model, and for the 3D model they usedResNet, DenseNet, NASLung and the 3D CNN proposed in their paper. Their 3D CNN was the best performance achieved and you can see in Table 4 with an accuracy of 0.751, an mAP of 0.820 and 0.812 respectively. The performances of NASLung and 3D DenseNet were not quite as good as their proposed CNN. However, the number of parameters required in those models are a lot higher than their proposed 3D CNN, meaning their CNN is much more effective.

I will try combine the Mean Teacher self training network with consistency Regularisation, as I can apply this in the MLC task of nodule signs. I will also be able to compare my model to this paper and hopefully can achieve somewhat similar results to many of the other models.

Why Segmentation Over MLC for This Project

Why Segmentation Over MLC While MLC approaches enable simultaneous sign detection, this project focuses on voxel-level segmentation because it better matches both clinical needs and the LUNA16 dataset characteristics. Precise boundary delineation gives accurate volume measurements needed for Lung-RADS risk stratification, whereas MLC only provides binary sign presence. The LUNA16 annotations with 3D nodule coordinates naturally generate sphere-based segmentation masks rather than requiring unavailable sign labels. Segmentation also learns rich spatial features transferable to volume analysis and texture tasks, while MLC remains sign-specific. Finally, Mean Teacher consistency loss works naturally on pixel/voxel predictions, avoiding the complex multi-label consistency design needed for MLC. Segmentation thus provides a more robust foundation for semi-supervised learning while directly enabling clinically critical nodule volumetry.

Table 4: Performance comparison of the MLC models trained from scratch

Model	Parameter	Accuracy	mAP	F1-micro	F1-macro	Hamming loss
Local_Global (2D)	183.99 k	0.739	0.838	0.809	0.793	0.261
ResNet-34 (3D)	63,472.71 k	0.719	0.828	0.790	0.781	0.281
DenseNet-121 (3D)	11,248.77 k	0.748	0.845	0.817	0.808	0.252
NASLung (3D)	28,564.35 k	0.744	0.851	0.812	0.801	0.256
The proposed CNN	1,409.54 k	0.751*	0.860*	0.820*	0.812*	0.249*

*, the best performance in each column. MLC, multilabel classification; mAP, mean average precision; 2D, two-dimensional; 3D, three-dimensional; CNN, convolutional neural network; k, thousand.

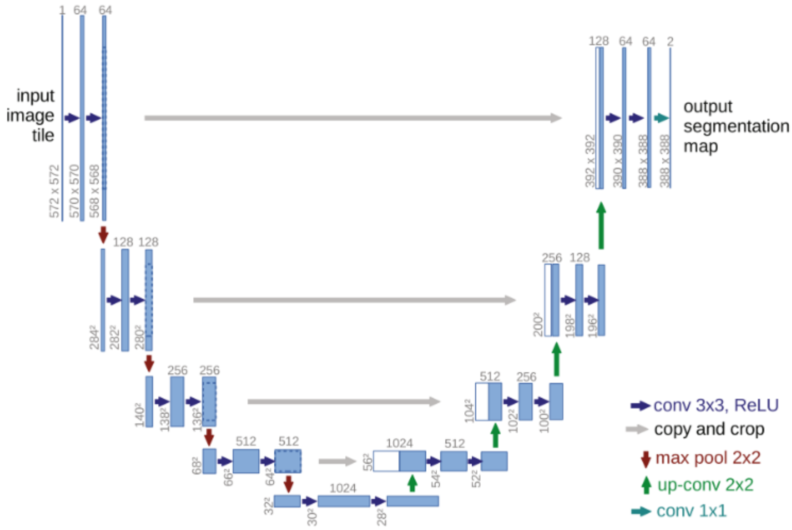


Figure 1: This is the U-net structure where each blue box corresponds to a multi-channel feature map. The number at the top of the box denotes the number of channels. At the lower left edge of the box, the x-y-size is provided. The white boxes represent copied feature maps, and the arrows denote the different operations.

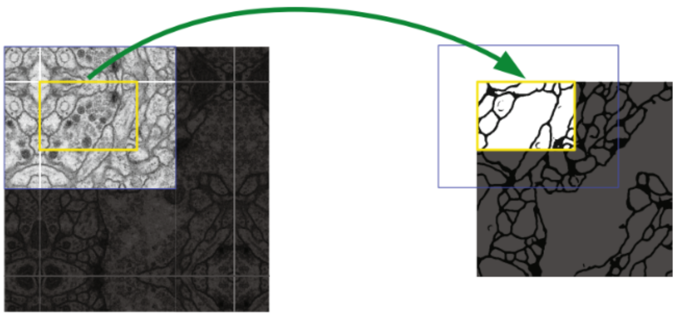


Figure 2: This is the overlap-tile strategy which allows for seamless segmentation of arbitrary large images (here segmentation of neuronal structures in EM stacks). In the yellow area there are the predictions of the segmentation, which requires image data within the blue area as input. The missing input data is extrapolated by mirroring.

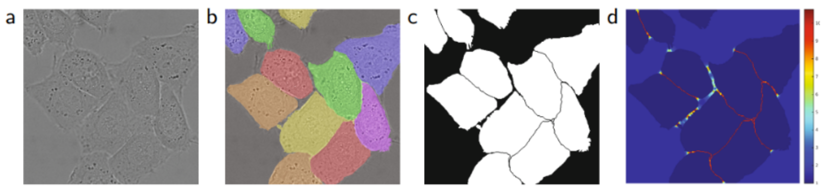


Figure 3: The HeLa cells on glass recorded with DIC (differential interference contrast) microscopy. The first image is (a) raw image. The second image is (b) overlay with ground truth segmentation where different instances of the HeLa cells are indicated by different colours. The third image is (c) generated segmentation mask (white: foreground, black: background). The fourth image is a (d) map with a pixel-wise loss weight that forces the network to learn the border pixels.

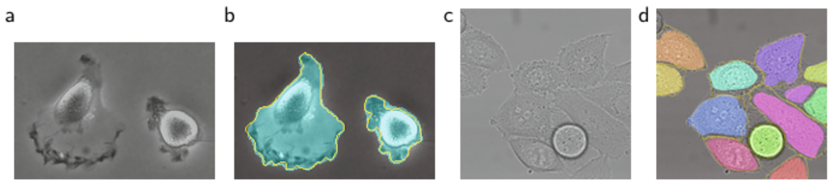


Figure 4: These are the Results on the ISBI cell tracking challenge. The first image is, (a) which is part of an input image of the "PhC-U373" data set. The second image is, (b) which is the Segmentation result (cyan mask) with manual ground truth (yellow border). The third image is, (c) which is an input image of the "DIC-HeLa" data set. The fourth image is, (d) which is a Segmentation result (random coloured masks) with manual ground truth (yellow border).

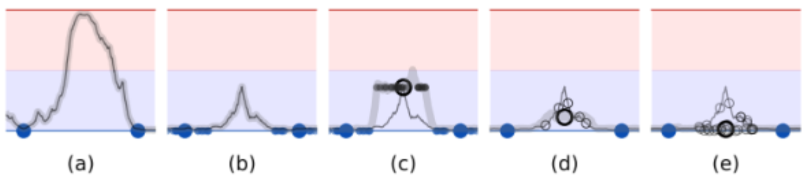


Figure 5: This is a sketch of a binary classification task with two labelled examples (large blue dots) and one unlabelled example, which demonstrates how the choice of the unlabelled target (black circle) affects the fitted function (gray curve). The first image is, (a) which is a model that has no Regularisation is free to fit any function that predicts the labelled training examples well. The second image is, (b) which is a model that is trained with noisy labelled data (small dots) learns to give consistent predictions around labelled data points. The third image is, (c) which shows how consistency to noise around unlabelled examples provides additional smoothing. For the clarity of illustration, the teacher model (gray curve) is first fitted to the labelled examples, and then left unchanged during the training of the student model. As well as this, for clarity, we will remove the small dots in figures d and e. The fourth image is, (d) which shows how noise on the teacher model will reduce the bias of the targets without additional training. The mean (large blue circle) of individual noisy targets (small blue circles) is the expected direction of stochastic gradient descent. The fifth image is, (e) which is an ensemble of models that gives an even better expected target. Both Temporal Ensembling and the Mean Teacher method use this approach.

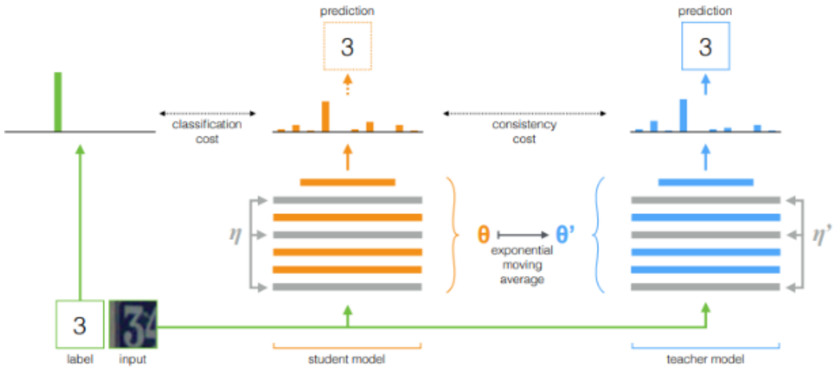


Figure 6: This is the Mean Teacher method. The figure depicts a training batch with a single labelled example where both the student and the teacher model evaluate the input applying noise (n,n') within their computation. We compare the softmax output of the student model to the one-hot label using classification cost and with the teacher output using consistency cost. The teacher model weights are updated as an exponential moving average of the student weights after the weights of the student model have been updated with gradient descent. Both the model outputs can be used for prediction, but at the end of the training the teacher prediction is more likely to be correct. A training step with an unlabelled example would be similar, except no classification cost would be applied.

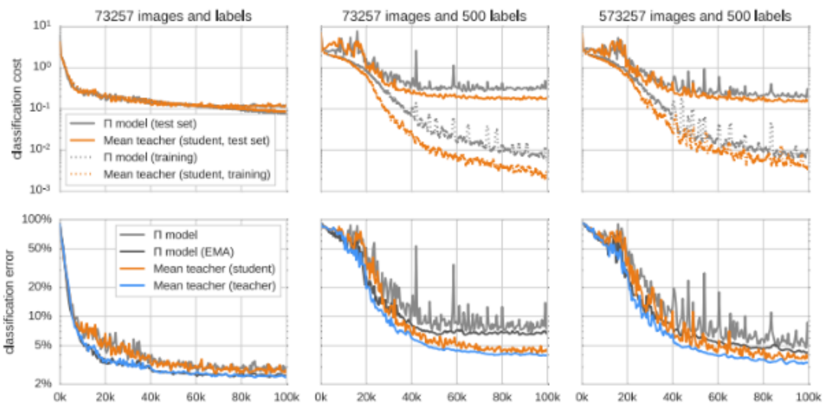


Figure 7: This is the smoothened classification cost (top) and classification error (bottom) of Mean Teacher and their baseline PIE model on SVHN over the first 100000 training steps. The training classification costs are measured using only labelled data in the upper row.

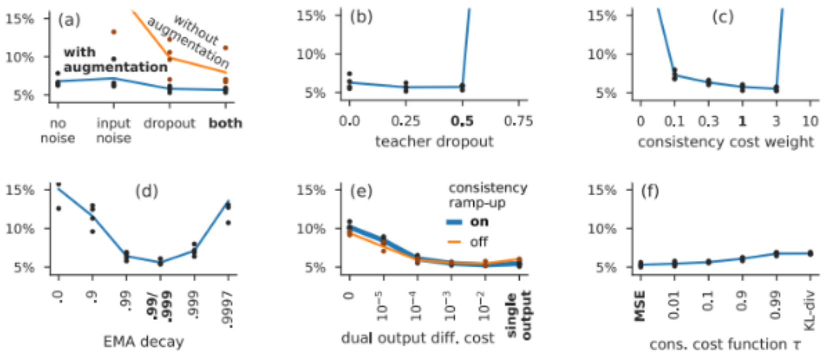


Figure 8: This is the validation error on 250-label SVHN over four runs per hyperparameter setting and their means. In each experiment, they varied one hyperparameter, and used evaluation run hyperparameters. The hyperparameter settings used in the evaluation runs are marked with the bolded font weight.

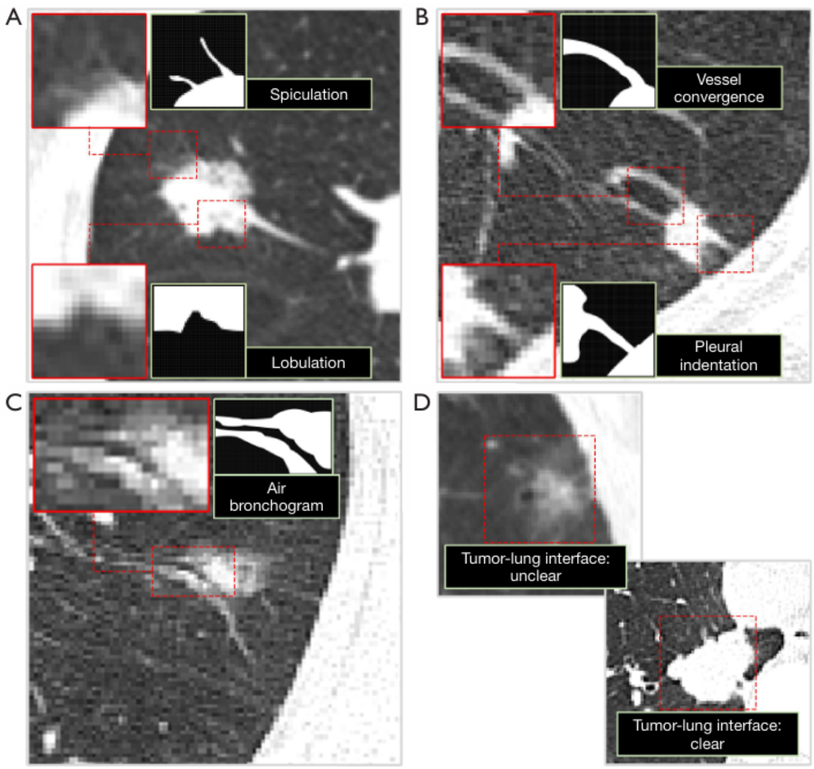


Figure 9: These are examples of common lung nodule CT signs. The first image is, (A) which shows Spiculation and lobulation; second image shows (B) vessel convergence, pleural indentation; third image shows (C) air bronchogram; and fourth image shows (D) tumor-lung interface. The nodule CT sign in the red dotted box is enlarged and displayed in the red solid line box in the upper left corner or lower left corner of the subfigure. The corresponding binary image is given in the light green box to better display the characteristics of different nodule signs. CT, computed tomography

Chapter 4

Methodology

4.1 Data description

The dataset I have used is LUNA16 (Lung Nodule Analysis 2016) from the grand-challenge.org which is a collection of thoracic computed tomography (CT) scans, and derived from the LIDC-IDRI lung cancer screening study. The formatting is several .zip and .csv files. I have chosen this dataset becaase the dataset provides precise 3D nodule locations and diameters, which I can convert into voxel-level segmentation labels for training and evaluation. Not all the scans will have labels or not many labels which produces scarcity and aligns perfectly with semi-supervised consistency regularisation (Mean Teacher). The dataset is large enough to train 3D networks like U-net and mean teacher and easily accessible as a student. Finally, there has been several deep learning papers that use LUNA16 on nodule detection that I have mentioned previously such as Setio et al. [2017] showing that its a good dataset.

These included 10 subsets of scans (Subset0.zip-subset9.zip) which are 888 CT scans in the format of .mhd and .raw/.zraw, and each scan is one patient series and identified by a SeriesInstanceUID string in the filename. I unzipped these files and stored them in the filepath data/scans/subset*/.

The annotations.csv file contains the reference standard nodule annotations for the "nodule detection" track. Each row is equal to one nodule where the seriesuid is the ID of the CT scan, the coordX, coordY, coordZ are the nodule centre in world coordinates (in mm) and diameter_mm is the nodule diameter in millimetres. There are 1186 nodules across the 888 scans. This is stored in data/

The candidates.csv contains the original list of nodule candidates for the ISBI 2016 false-positive reduction trackm where each row is equal to one candidate location: seriesuid, coordX, coordY, coordZ, class (1 = nodule, 0 = non-nodule). Usually superseded by candidates_V2.csv but kept for completeness. Located in the filepath data/.

The candidates_V2.csv which is inside candidates_V2.zip contains an updated and extended set of candidate locations for the false-positive reduction task. This was Produced by merging the original candidates with candidates from full CAD systems. It has higher sensitivity, so it detects 1166/1186 nodules with roughly 551k candidates.

The sampleSubmission.csv is an .csv file containing an example of a submission file format for the challenge. This shows the required columns (e.g. seriesuid, coordinates/probabilities) and CSV structure for evaluation. Located in the filepath data/.

The seg-lungs-LUNA16.zip is a Zip containing automatic lung segmentations (binary masks) for each CT scan. This has one .mhd mask per seriesuid, and is used to restrict processing to the lung region. Unzipped and located in the filepath data/seg-lungs.

The evaluationScript.zip contains the official evaluation code for LUNA16. These are scripts to compute FROC and other metrics on my predictions vs the reference annotations/candidates. Unzipped and located in data/scripts

All unzipped files are copied and located in data/raw_backups just in case.

4.2 Preprocessing

I loaded the relevant paths for the data so that they are ready for data preparation. Then I created and applied a lung mask to my data, which is a binary segmentation of the lungs that I used on each CT scan as a spatial filter. This only keeps lung tissue, so the voxels inside the lung are labelled 1 and everything else is labelled 0 (heart, chest wall, liver, and outside the body). This allows U-net/ Mean Teacher to focus on only the lung, meaning they are less likely to produce false positives and noise by not hallucinating nodules outside the lung. Also, this will save memory and as there is less volume to focus on which will in turn improve training speed.

```
# Lung mask
    seg_path = os.path.join(SEG_DIR, seriesuid + ".mhd")
    if not os.path.exists(seg_path):
        raise FileNotFoundError(f"Lung mask not found for {seriesuid}: {
            seg_path}")
    lung_img = sitk.ReadImage(seg_path)
    lung_arr = sitk.GetArrayFromImage(lung_img) > 0
```

After that, I used CT windowing, which means that only part of the image will be saved to reduce extreme outliers, as the voxels in Hounsfield Units (HU) and are roughly from -1000 to +1000. The useful range is between -1000 and +600 HU, so my algorithm focuses on that range to capture all relevant pulmonary structures. Then I applied normalisation which allows a voxel at -1200 HU to become 0.0, and a voxel at 600 HU to become 1.0, allowing a normalised floating volume which is what the U-net algorithm expects to take as input. Also ensures that the model's weights remain stable during gradient descent, as high-magnitude input values can lead to vanishing or exploding gradients.

```
# Window to lung HU range and scale to [0,1]
    # Typical lung window ~[-1200, 600] HU[web:46]
    ct_arr = np.clip(ct_arr, -1200, 600)
    ct_arr = (ct_arr + 1200.0) / 1800.0   # -1200 -> 0, 600 -> 1
```

Due to the computer seeing images as an array of voxels, I have used the TransformPhysicalPointToIndex function from the SimpleITK library to convert LUNA16 annotations from "World Space" (millimetres relative to a scanner origin) to voxels. This is critical to ensure that the ground truth mask would not align with the actual nodule in the image. After this, I converted the point annotations into 3D binary spheres based on the actual nodule diameter, to provide a stronger learning signal. This volumetric target helps the Mean Teacher framework recognise the shape and boundary of the nodule, rather than just a single pixel, allowing for a more accurate result.

```
# SimpleITK helper to get Voxel Index (x, y, z)
        voxel_idx = ct_img.TransformPhysicalPointToIndex(world_coord)
        v_x, v_y, v_z = voxel_idx
```

Finally, my algorithm sorted the scans into labelled and unlabelled folders in preparation for the U-net algorithm using the seriesuid column in the annotations.csv file. Of the 888 scans, there were a total of 300 annotated nodules, and they were designated as labelled data, with the remaining only 270 scans and masks containing no annotations used as unlabelled.

4.3 Dataset and Batching

4.3.1 3D Patching

Due to high dimensionality of thoracic CT scans, which often reach 512 x 512 x Z voxels, processing full volumes is often computationally costly. Even on high-performance deep learning computers, the intermediate feature maps generated by 3D convolutions at each layer would exceed the available Video RAM (VRAM).

To help with this GPU memory bottleneck (to ensure computational stability and prevent Out-of-Memory (OOM) errors.), I implemented a Fixed-Size Cuboid strategy, in which I extracted 643 from the global volumes.

This is so that the data can transition from Global Volume to Local Patches to allows my data to be processed at a high resolution by the model which maintaining a reasonable batch size.In terms of voxel spacing, the LUNA16 scans have different slice thicknesses, so my dataset.py class will ensure that a 64|3 patch covers a consistent uniform volume in millimetres.

4.3.2 Sampling Strategy

```
# Randomly select the top-left corner of the patch
        start_z = np.random.randint(0, z - pz)
        start_y = np.random.randint(0, y - py)
        start_x = np.random.randint(0, x - px)
```

I used a stochastic approach for the sampling logic, where I started off by extracting a random 3D patch and then continued to select a random top left-coordinate on the batch for each training iteration. This is for the labelled pool so that the network is forced to learn translation invariance,allowing U-net to detect a nodule regardless of its position within the input cuboid. This is compared with the same spot on the image each time, where the model may develop a spatial bias. For the unlabelled pool, this allows the model to learn a generalised representation of 'normal' lung anatomy, which strengthens the mean teacher's ability to distinguish background noise from lung features.

4.3.3 Tensor formatting and patch preparation

```
# Add channel dimension (C, Z, Y, X) for PyTorch 3D Conv
        img_patch = torch.from_numpy(img_patch).unsqueeze(0).float()
        mask_patch = torch.from_numpy(mask_patch).unsqueeze(0).float()
```

From the above code, the input volumes are formatted as 4D tensors with the structure (C, Z, Y, X) that satisfies the requirements of the PyTorch 3D convolutional layers. Whereas the segmentation masks are binary (0,1), they are cast to float32 precision. Therefore, this allows for the calculation of the Dice Loss and Mean Squared Error (Consistency Loss), which require floating-point gradients to backpropagate through the Student network during the semi-supervised training phase. Ensure that the student model ”learns” from the soft map probability maps generated by the teacher model.

4.4 Model Architecture

4.4.1 Baseline U-Net architecture

Figure 10 illustrates the full supervised baseline training pipeline. Labelled data flows through the 3D ResUNet, supervised loss is computed, and gradients are backpropagated to update the model weights.

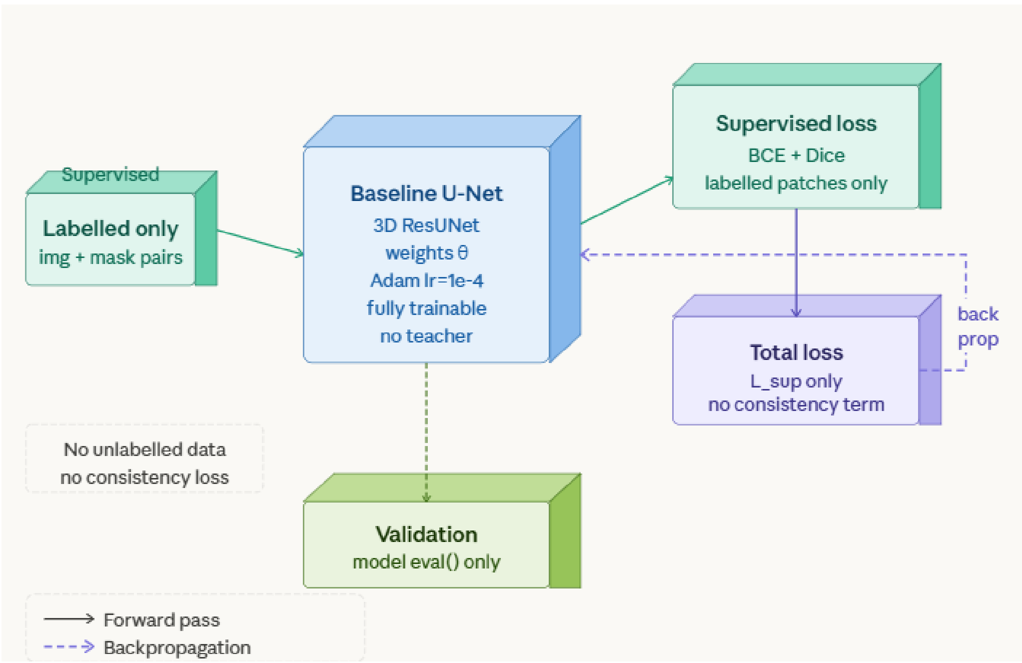


Figure 10: This is the Supervised Baseline training pipeline. Labelled image and mask pairs pass through the 3D ResUNet, producing logits that are compared against the ground truth using BCE + Dice loss. Backpropagation updates the model weights directly. No unlabelled data or consistency regularisation is used.

Residual

My baseline is a modified 3D U-net that incorporates Residual Learning to enhance feature extraction from complex volumetric lung data. First, I developed a ResidualBlock3D class. From He et al. [2015] we can use the Residual block to let the network learn the residual mapping ($F(x)$) rather than the full underlying mapping ($H(x)$).

In the code below, I have used a shortcut, x, which allows us to pass through a useless layer that may otherwise distort the data, pushing the input x directly to the next layer. This code performs the following:

$$y = \mathrm{ReLU}(F(x) + \mathrm{shortcut}(x))$$

This allows the gradient to not be diminished by repeated matrix multiplications and still flow through the addition operator to earlier layers. In context of LUNA16, this means the sharp edges of a 5mm nodule are preserved even after the data has passed through 20+ convolutional layers.

```
self.shortcut = nn.Sequential()
        if in_ch != out_ch:
            self.shortcut = nn.Sequential(
                nn.Conv3d(in_ch, out_ch, kernel_size=1),
                nn.BatchNorm3d(out_ch)
```

However, due to the number of input channels (in_ch) and output channels (out_ch) not matching such as when the encoder transitions from 32 to 64 filters, a mathematical challenge arises. Hence, I implemented the shortcut projection and used a 1 x 1 x 1 convolution. A 1x1x1 kernel changes the depth of the data without altering its spatial dimensions, allowing the ”highway” path to provide clean, un-distorted data of the original anatomy to the deeper layers of the network.
I have used:

```
                nn.BatchNorm3d(out_ch)
```

to ensure the scale of the ”shortcut” data matches the scale of the ”learned” data. This means if one path has much higher values than the other, the addition

```
                out += residual
```

would be domination by one side, making the other side useless. So normalising both paths helps keep them in the same numerical range.

The model architecture was implemented by adapting the residual block logic from Yorke [2024], while the overall 3D U-Net structural framework was based on the implementation described by codegenes [2024].

Encoder

After completion of the residual function, I started with the encoder of the U-net network.

```
        self.enc1 = ResidualBlock3D(in_channels, base_ch)
        self.pool1 = nn.MaxPool3d(2)
```

The first encoder uses the ”ResidualBlock3D” class to transform incoming channels (raw Hounsfield Unit intensities) into a single channel into 32 feature maps. This first level acts as an ”edge” detector. It identifies sharp contrast between the air and the lung tissue or nodule modules. The Residual class is used to ensure fine textures of the lung parenchyma are not lost before the first downsampling occurs.

The second and third level of encoding is where the spatial resolution drops but the ”Feature Depth” increases. Everytime ”nn.MaxPool3d(2)” is called, the volume is reduced by a factor of 8 ($2 \times 2 \times 2$). To compensate for this loss of spatial information, the number of channels doubles. So at level 3, the model is no longer looking at individual voxels, it is looking at anatomical structures. A $3 \times 3 \times 3$ kernel at this depth covers a much larger physical area of the lung than it did at Level 1. This allows the model to recognise that a round object is a ”nodule” because of its relationship to nearby blood vessels or the pleural surface, rather than just its colour.

Bottleneck

After encoding, I then started on the bottleneck, which is the bridge between the encoder and the decoder.

```
        self.bottleneck = ResidualBlock3D(base_ch * 4, base_ch * 8)
```

From above you can see that this is the only level where no pooling occurs immediately after the block. The feature depth is at its highest (256), but its resolution is at its lowest (8^3). This 256-channel vector contains the most abstract semantic information and so represents the ”essence” of the 64^3 patch. In my Mean Teacher framework, the bottleneck is the part of the model that learns the most robust, noise-resistant features of the lung anatomy.

Decoder

Then I began the decoder block. This is responsible for taking the abstract semantic ”essence” from the bottleneck and projecting it back into a 64^3 voxel-wise segmentation mask.

```
         # Decoder
        self.up3 = nn.ConvTranspose3d(base_ch * 8, base_ch * 4, kernel_size
            =2, stride=2)
        self.dec3 = ResidualBlock3D(base_ch * 8, base_ch * 4) # 4 (up) + 4
            (skip)
```

This path utilises nn.ConvTranspose3d. This is a learnable up-sampling layer compared to normal standard linear interpolation, which just stretches the image. This allows the network to learn the most effective weights for ”re-inflating” the feature maps. This is key for LUNA16, where nodule boiundaries are often jagged and complex, the model is now able to learn to fill in the spatial gaps rather than just blurring the edges, making it easier for the Mean Teacher model.

Forward Loop

In the forward loop, encoder features are concatenated with up-sampled decoder features.

```
d3 = self.up3(b)
d3 = self.dec3(torch.cat([e3, d3], dim=1))
```

By using torch.cat, I am able to ”bridge” the high-resolution spatial information from the encoder (e3) directly to the decoder. The up-sampled features (d3) provide the semantic knowledge that a nodule exists, to tell you what is there and the encoder features (e3) provide the high resolution location and boundary details that were lost during max-pooling, to tell you where.

As we can see in the code, self.dec3 receives 256 channels (base_ch * 8) which is the sum of the up-sampled features (128) and the skipped features (128). Then I used ResidualBlock3D to combine both channels to produce an anatomically correct mask.

Final Voxel Classification

In the final line of the LUNA16_UNet init, the final layer of the network collapses the 32 feature maps from the last decoder level into a single output channel.

```
self.final = nn.Conv3d(base_ch, out_channels, kernel_size=1)
```

I used a convolution $1 \times 1 \times 1$ here to perform voxel-wise classification. This layer uses Conv3d to assign a probability to every voxel in the 64^3 patch, which indicates the likelihood of that voxel being part of the voxels included in the primary nodule.

4.4.2 Mean Teacher semi-supervised framework

Figure 11 illustrates the full Mean Teacher training pipeline, contrasting with the simpler baseline in Figure 10.

The main challenge of the LUNA16 dataset is the small amount of labelled data compared to unlabelled data, where labelled data can only be used for image segmentation. This is why I have implemented the Mean Teacher framework which is a semi-supervised architecture that uses Consistency Regularisation. The framework moves beyond standard supervised learning by requiring the model to produce stable, consistent predictions across both labelled and unlabelled data.

The framework consists of two identical 3D U-Net architectures: the Student (f_θ) and the Teacher ($f_{\theta'}$).

The Student (f_θ) is a model that is the active learner. It is updated via backpropagation using a dual-objective loss function: a supervised loss on labelled patches and a consistency loss on all patches. Where as the Teacher ($f_{\theta'}$) acts as a "temporal ensemble" of the Student. This model does not learn from gradients, as you can see:

```
for param in self.teacher.parameters():

            param.requires_grad = False
```

Instead, its role is to provide a "stable target" for the Student to aim for when processing unlabelled data. The teacher only changes through the smoothed evolution of the student. Confirmation bias would occur if the Teacher was able to learn from gradients. Also, if both models were updated based on the same loss, they would both converge to a "trivial" solution to satisfy the consistency requirement.

I initialised the Teacher using a Deep Copy of the Student:

```
self.teacher = copy.deepcopy(model)
```

This happened before training to ensure both the student and the teacher being from the same baseline, which prevents the Student from being forced to 'chase' a randomly

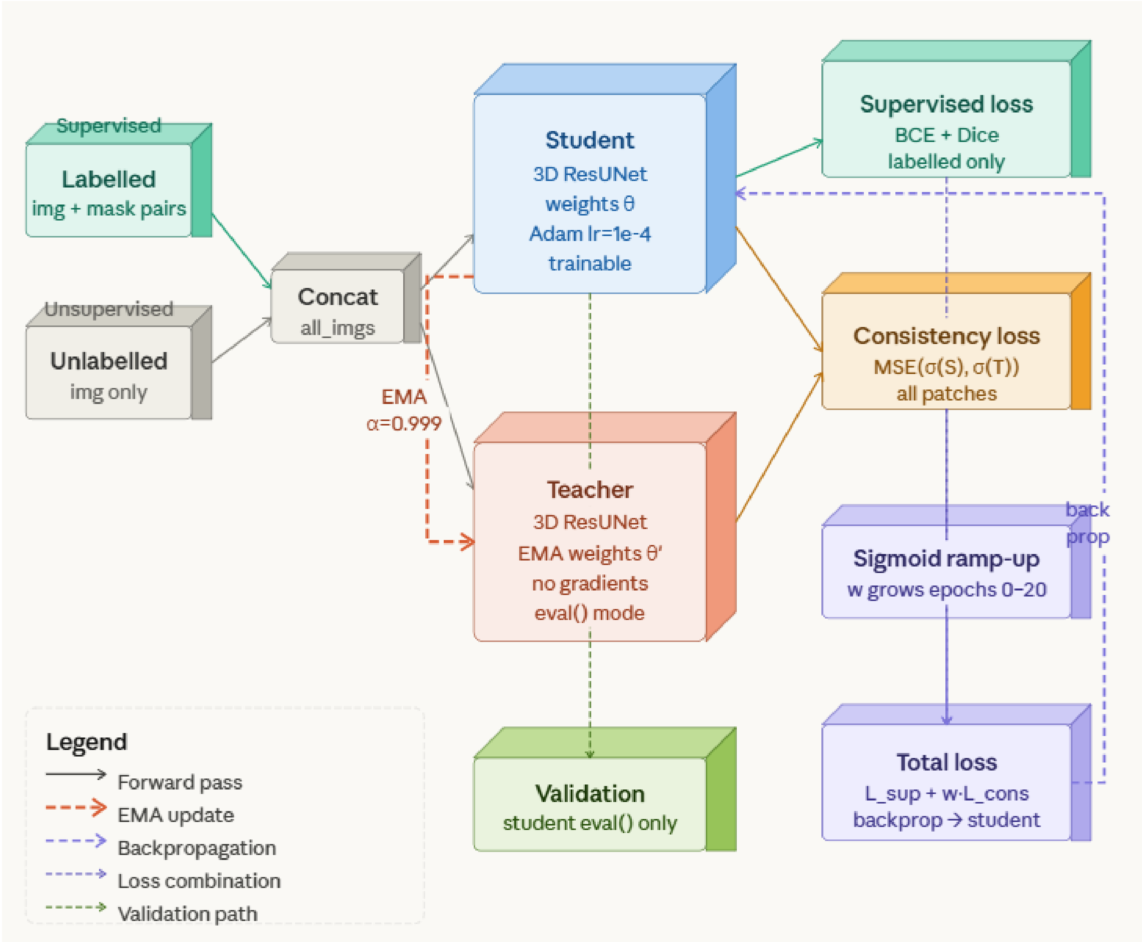


Figure 11: Mean Teacher semi-supervised training pipeline. Labelled and unlabelled patches are concatenated and passed through the student and teacher networks. The student is updated by backpropagation using the combined supervised and consistency losses, weighted by a sigmoid ramp-up schedule. The teacher is updated only through an exponential moving average of the student weights ($\alpha = 0.999$) and remains in evaluation mode throughout training.

initialised Teacher. This would lead to divergence and poor performance in the early epochs.

Exponential Moving Average (EMA)

The Exponential Moving Average (EMA) of the Students Weight over time is denoted as the "intelligence" of the Teacher.

$$\theta'_t = \alpha\theta'_{t-1} + (1 - \alpha)\theta_t \tag{4.1}$$

Here α is the smoothing coefficient (set to 0.999). I have used this logic in the update_weights method.
This uses Temporal Smoothing when we set $\alpha = 0.999$, the Teacher will retain 99.9% of its previous state and only incorporates 0.1% of the Student's current weights. This allows the Teacher to be highly resistant to the "noisy" updates that occur when the student encounters a particularly difficult 3D lung patch. This in turn reduces updating the Teacher incorrectly, strengthening my results.

Also, in 3D imaging the batch sizes are often too small (e.g. 4 or 8) due to VRAM limits, and small batches produce high-variance gradients. The EMA is able to act as a low-pass filter, smoothing out these fluctuations so that the Teacher represents what the Students learning trajectory will be.

The Mean Teacher framework operates so that even if a scan is perturbed by random cropping in the dataset.py, the predicted nodule should remain consistent. As we force the student to match the Teachers output on the unlabelled data, the network is able to learn anatomical features that are invariant to noise. In LUNA16, this is necessary to ensure that nodules are not confused with blood vessels or scars. We use Consistency Regularisation to ensure the model only ”believes” in features that appear consistently across multiple views and model states.

4.4.3 Consistency loss design

The Mean Teacher framework is trained by the total objective function (L_{total}) which is a composition of two distinct loss terms: a Supervised Loss (L_{sup}) and a Consistency Loss (L_{con}). This allows the model to learn from the ground-truth masks while simultaneously regularising its behaviour on the vast pool of unlabelled LUNA16 scans.

Supervised Loss: BCE + Dice Coefficient

I used Dice Loss for the labelled portion of the dataset. Due to Dice Loss specifically addresses the class imbalance problem, it is seen as superior to standard Cross-Entropy.In a 64^3 patch, a nodule may only occupy 0.1% of the total voxels. The Dice Loss will only focus on the overlap between the prediction and the ground truth, preventing the background voxels from overwhelming the gradient.

I also used Binary Cross-Entropy (BCE) added too:

$$L_{sup} = L_{BCE} + L_{Dice}$$

This is because BCE provides a stable voxel-wise gradient signal throughout training, particularly in early epochs, when predictions are far from the ground truth. The combination ensures both pixel-level accuracy and overlap-based optimisation. The combination produces more stable convergence than either loss alone, as demonstrated in the medical image segmentation literature Rajput [2021]. The code for my loss supervised is:

```
loss_sup = F.binary_cross_entropy_with_logits(lab_logits, mask_lab)
+ dice_loss(lab_logits, mask_lab)
```

Consistency Regularization: Mean Squared Error (MSE)

On the other hand, the consistency loss is calculated by comparing and calculating the squared.

Euclidean distance between the probability maps generated by the Student (P_{stud}) and the Teacher (P_{teach}) for the same input patch. The Mean Squared Error (MSE) was used as the distance metric:

$$L_{con} = \frac{1}{N}\sum(P_{stud} - P_{teach})^2$$

Total loss

The total training loss for the mean teacher programme is therefore defined as:

$$L_{total} = L_{sup} + w_{cons} \cdot L_{con}$$

where w_{cons} is the sigmoid ramp-up weight described in Section 4.4.4. During early training when $w_{cons} \approx 0$, the model will almost train entirely on the supervised signal. As w_{cons} approaches 1, the consistency term contributes equally, which encourages the student to produce predictions consistent with the teacher across both labelled and unlabelled patches.

4.4.4 Training setup and hyperparameters

The consistency loss is computed using PyTorch's built-in `F.mse_loss` function, which automatically handles the 5D tensors ($Batch, Channel, Z, Y, X$), calculate the square of every individual voxel, and then "reduces" them into a single scalar value. This is also achievable for the GPU and handles floating-point precision more efficiently than a manual Python loop due to being more cumerically stable.
To ensure stability of semi-supervised learning, I implemented a Sigmoid Ramp-up schedule for consistency weight (w_{cons}).

```
w_cons = sigmoid_rampup(epoch, rampup_length=20)total_loss = loss_sup + (
    w_cons * loss_cons)
```

This schedule is critical because, during the initial 20 epochs, the Teacher model has not yet developed a thorough representation of the lung anatomy. Using $w_{cons} \approx 0$, the Student model is allowed at the start to establish a baseline using expert-labelled nodules. As the ramp-up progresses, the model shifts its focus toward the larger unlabelled dataset, learning from the Teacher's evolving data to improve its segmentation boundaries.

During training, I created two training scripts: a supervised baseline (Baseline_train.py) and the semi-supervised Mean Teacher framework (Mean_teacher_train.py). Both scripts share an identical pipeline, the same 80/20 labelled train/validation split with a fixed random seed of 42, and the same 3D ResUNet architecture (which i established earlier) with identical hyperparameters of batch size of 2, learning rate of 1×10^{-4}, and 100 training epochs. The data pre processing was the exact same too to ensure that any difference in performance of the two models is attributable to the semi-supervised model rather than differences in data preparation. The baseline script is exclusively allowed to use supervised loss to train on labelled data, whereas the Mean Teacher is allowed to also use the unlabelled data through consistency regularisation, utilizing Mean Squared Error (MSE) between the student and teacher probability maps as the consistency loss, following the method proposed by Tarvainen and Valpola [2018].

4.4.5 Comparison of training setups

Table 5 summarises the key differences between the two training programmes. Both models share the same 3D ResUNet architecture, optimiser, learning rate, batch size, and random seed to ensure a controlled comparison. The differences are the use of the unlabelled data loader folder, the consistency loss term, the sigmoid ramp-up schedule, and the EMA-updated teacher network in the Mean Teacher programme. Any improvement in segmentation performance will be due to the semi-supervised consistency regularisation framework.

Table 5: Comparing the two frameworks set up

Component	**Baseline**	**Mean Teacher**
Architecture	3D ResUNet	3D ResUNet (Identical)
Labelled data	240 training scans	240 training scans
Unlabelled data	None	135 scans
Loss function	BCE + Dice	BCE + Dice + w · MSE
Consistency weight	None	Sigmoid ramp-up (0→1, epochs 0–20)
Teacher network	None	EMA copy, =0.999
Optimiser	Adam, lr=1e-4	Adam, lr=1e-4
Epochs	100	100
Batch size	2	2

4.4.6 Parameter Opitmization

For weight updates, I used the Adam Optimizer with a learning rate of 1×10^{-4}. I chose Adam due to its adaptive moment estimation Kingma and Ba [2017], which is particularly effective in 3D medical imaging where gradients can be sparse and non-convex.

```
optimizer = optim.Adam(student_net.parameters(), lr=lr)

... inside loop ...
total_loss.backward()
optimizer.step()
mt_manager.update_teacher()
```

Immediately following the Student's optimisation step, the update_teacher() method is called to perform the EMA weight update. By constantly updating the Teacher to reflect the Student's state, we maintain the consistency needed for high-quality semi-supervised results. Also, Baseline.py uses the same Adams optimiser, but without the update_teacher() call.

Chapter 5

Experiments and Results

5.1 Experimental Design

All experiments used the same hardware which was connected to Loughborough University remote Linux workstation that had a NVIDIA RTX 4000 Ada Generation GPU (20GB VRAM). I connected to the computers using SSH. Although this allowed me to actually run the 3D volumetric training programme without hitting out-of-memory errors, the university machines operated under scheduled nightly shutdowns at approximately 3am. To mitigate this, I had to use a resumed training system based on checkpoints. This saved the best-performing model weights after each epoch and allowed me to resume from the latest checkpoint upon reconnection. In some cases, promising training runs were interrupted before reaching new performance peaks due to either shutdown or connection issues, which may have introduced minor variability in the final results.

This is also the reason why I limited my data size to a subset of the entire LUNA16 corpus. Due to remote connection constraints and storage limitations on the university workstation, 300 labelled scans (D_L) and 135 unlabelled scans (D_U) were used instead of the entire 888 scans. Although the labelled set provided the primary supervised signal, the 135 unlabelled scans were integrated with the Mean Teacher framework to evaluate whether consistency regularisation could further refine a model already trained on a substantial labelled corpus.This controlled design means that any difference in performance is attributable to the semi-supervised framework rather than data variation.

To train my data, I used the 80/20 split, which is 80% of the data for training the data and 20% to validate my data. I also used a Pytorch 42 fixed seed to ensure identical data splits in both experiments. The validation data are used with the student set in evaluation mode and the torch set without gradient, which improves time and saves VRAM.

An important limitation of this evaluation is that the validation set used during the selection training (saving the best checkpoint) was also used for the final evaluation. This may produce slightly optimistic absolute metrics; however, since both models were evaluated on the same validation set under identical conditions, the relative comparison between them remains valid.

5.2 Evaluation Metrics

I used 4 standard medical image segmentation metrics to evaluate both models. From the systematic review of deep learning methods for the segmentation of lung cancer nodules Vos et al. [2024], the standard performance metrics for the segmentation of nodules are the Dice Similarity Coefficient (DSC), Hausdorff Distance and Intersection over Union (Jaccard Index). Also from the evaluation protocol established in semi-supervised medical image segmentation Yu et al. [2019a], the four metrics are reported Dice, Jaccard, HD95, and ASD. I decided to use Dice, Jaccard, as they measure volumetric overlap whilst HD95, and ASD capture boundary accuracy, together providing an assessment of both region-level and surface-level segmentation quality

The **Dice Similarity Coefficient (DSC)** measures the volumetric overlap between the predicted mask P and the ground truth G:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}$$

A score of 1.0 indicates perfect overlap and a score of 0 indicates no overlap.
The **Jaccard Index** (Intersection over Union) is a stricter overlap measure:

$$\text{Jaccard} = \frac{|P \cap G|}{|P \cup G|}$$

The **95th Percentile Hausdorff Distance (HD95)** measures boundary accuracy by computing the 95th percentile of all surface-to-surface distances between P and G. So lower values indicate better boundary delineation.

The **Average Surface Distance (ASD)** measures the mean distance between the predicted and ground truth surfaces. Like HD95, lower values indicate better boundary quality. Both HD95 and ASD are reported in voxels; to convert to millimetres, we multiply by the approximate LUNA16 voxel spacing of 0.7mm.

5.3 Baseline Performance

Table 6: Quantitative comparison of Supervised Baseline and Mean Teacher on the LUNA16 validation set

Model	**Dice**	**Jaccard**	**HD95 (vox)**	**ASD (vox)**
Supervised Baseline	0.7827 ± 0.2919	0.7189 ± 0.3236	5.12 ± 6.76	1.55 ± 1.03
Mean Teacher (SSL)	0.8177 ± 0.2682	0.7561 ± 0.2915	4.05 ± 6.37	1.09 ± 0.84

The supervised baseline achieved a mean Dice of 0.7827 in 61 validation scans. We can use figure 12 to see the validation Dice per epoch during training. The Baseline shows rapid development during initial learning, reaching approximately 0.35 during the first 10 epochs, followed by a period of high variance before gradual improvement thereafter. During training, considerable epoch-to-epoch variance can be observed throughout training, which is due to the random patch sampling used during validation. This is where each epoch evaluates on a different random subset of patches rather than just the volume, which introduces noise into the epoch score. The best checkpoint achieved a validation Dice of 0.6585 during epoch 84, which was a large increase from the previous highest validation Dice in epoch 71 that scored 0.6171.

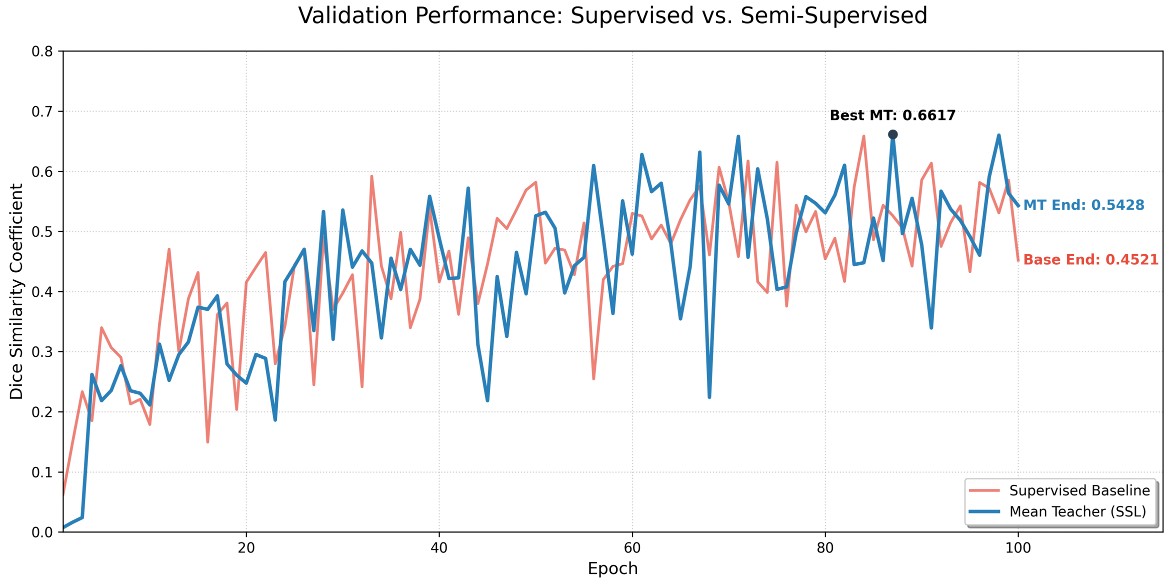


Figure 12: Validation Dice Similarity Coefficient per epoch for the Supervised Baseline (red) and Mean Teacher (blue) over 100 epochs. The marked point indicates the Mean Teacher best checkpoint at epoch 87 (Dice = 0.6617). Both models converge from near-zero within the first five epochs, with Mean Teacher demonstrating a consistently higher upper envelope from epoch 56 onward. High epoch-to-epoch variance in both curves is due to the random patch sampling strategy during validation, where each epoch evaluates on a different random subset of patches rather than fixed full-volume inference. The final epoch values (MT: 0.5428, Baseline: 0.4521) are lower than the best checkpoints due to this variance, which is why best-checkpoint evaluation is used for the quantitative results in Table 6

5.4 Semi-supervised Model Performance

The Mean Teacher framework achieved a mean Dice of 0.8177, a Jaccard of 0.7561, an HD95 of 4.05 voxels, and an ASD of 1.09 voxels. Across all four metrics, the Mean Teacher outperformed the supervised baseline. The Mean Teacher achieved a Dice of 0.8177 compared to the baseline's 0.7827, a Jaccard of 0.7561 versus 0.7189, an HD95 of 4.05 versus 5.12 voxels, with the most practically significant improvement being in ASD (1.55 $\rightarrow$ 1.09 voxels, a 29.7% reduction), indicating significantly better boundary quality as well as improved volumetric overlap. This is shown in Figure 13.

The mean teacher also showed a lower standard deviation in Dice (0.2682 vs 0.2919), suggesting that consistency regularisation produces more stable predictions across the validation set.

Figure 14 confirms this, with the Mean Teacher having not only a higher mean Dice

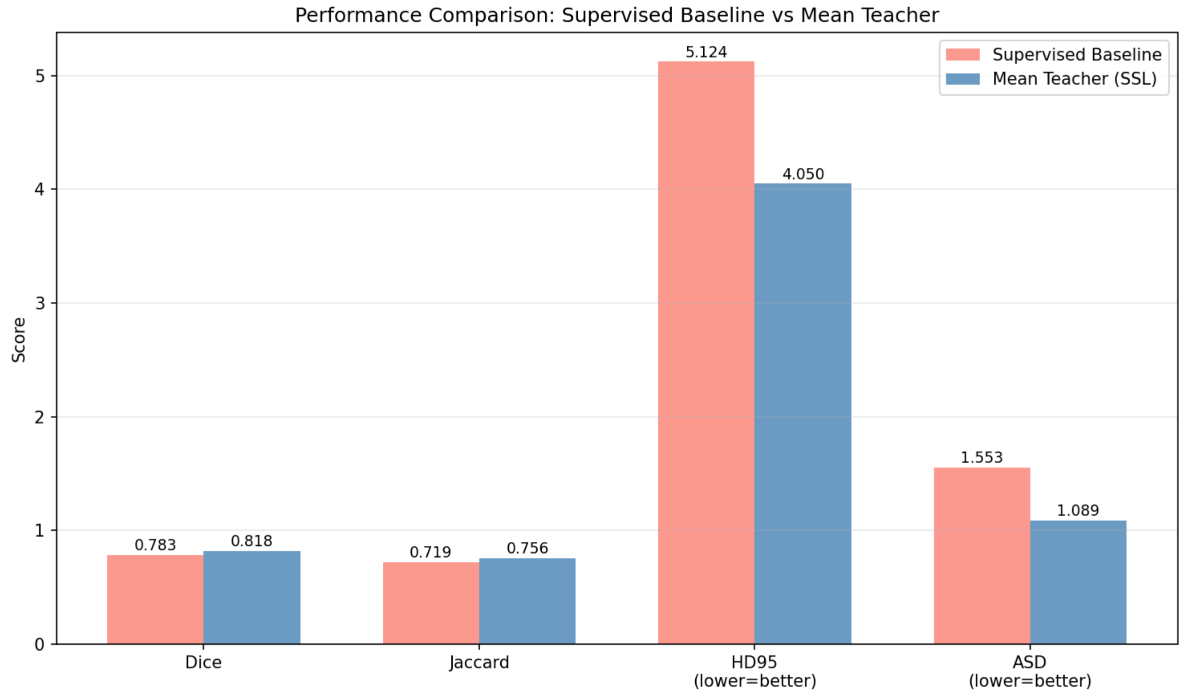


Figure 13: Comparison of Baseline and Mean Teacher performance. The Mean Teacher model consistently outperforms the supervised baseline across all metrics, most notably achieving a 29.7% reduction in Average Surface Distance (ASD) from 1.55 to 1.09 voxels.

but also Interquartile range being tighter and the number of low-performing outlier scans being reduced compared to the baseline, showing a better performance across the Validation set. Figure 15 shows the per-scan Dice for both models plotted against each other during training. Points above the diagonal dotted line show scans in which the Mean Teacher outperformed the baseline, and points below indicate the reverse. We can see the majority non-trivial scans (Dice < 1.0) lie above the diagonal, which confirms that Mean Teacher's advantage is consistent across difficult cases rather than driven by a small number of outliers. The cluster of points at (1.0, 1.0) represents easy scans where both models achieved perfect overlap.

The scatter plot confirms that Mean Teacher's advantage is across the full range of scans rather than being driven by a small number of outliers. The majority of partial-overlap scans (Dice < 1.0) lie above the diagonal, with the most notable improvements occurring on scans where the baseline scored between 0.4 and 0.6, which is where the difficult cases where consistency regularisation on unlabelled data is expected to provide the most benefits.

5.5 Qualitative Segmentation Examples

Figure 16 presents a qualitative comparison of both models on the same representative LUNA16 validation scan (scan_730). The Supervised Baseline result is taken from the final epoch 100 checkpoint (Validation Dice = 0.4521), whilst the Mean Teacher result is taken from its best checkpoint (epoch 87, Validation Dice = 0.6617). For this specific scan, the Supervised Baseline achieved a Dice of 0.4521 and the Mean Teacher achieved

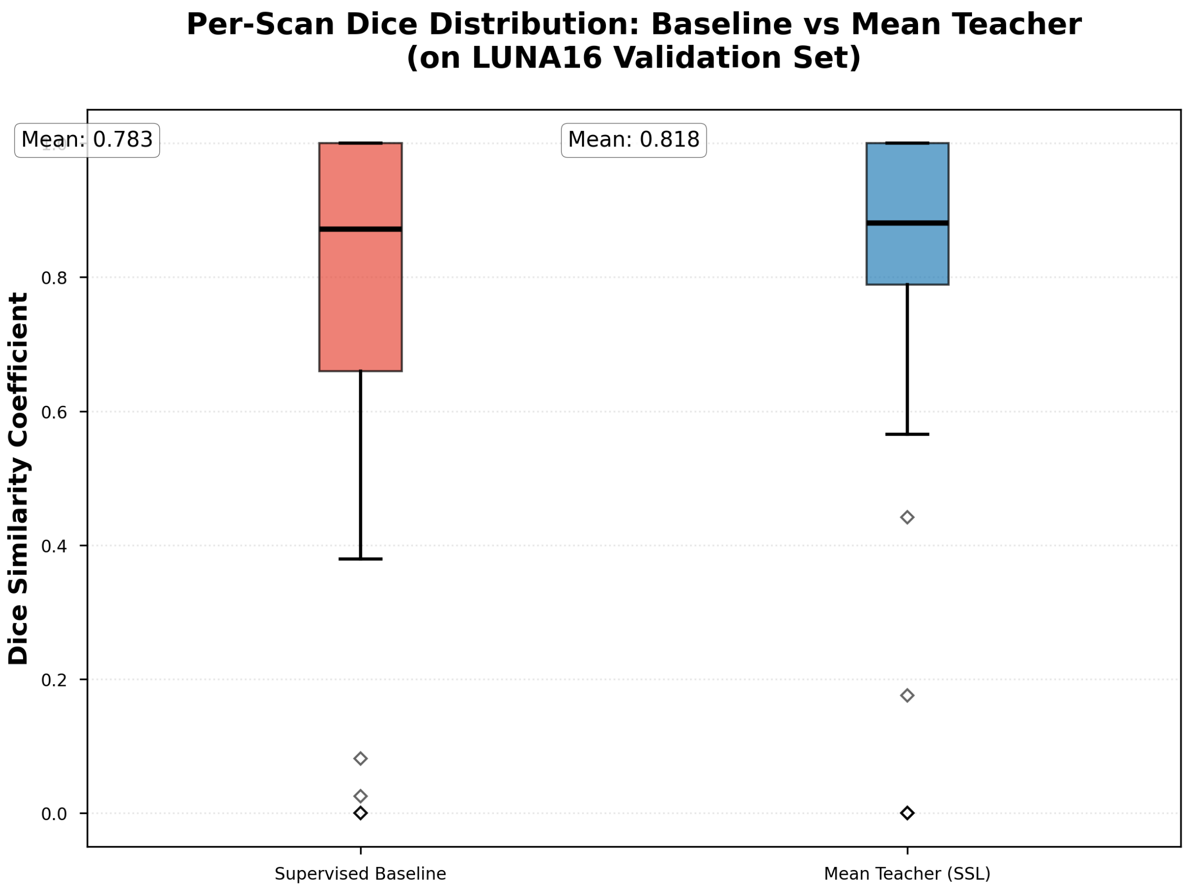


Figure 14: Per-scan Dice distribution for both models on the 61-scan validation set. Mean Teacher (blue) shows a higher mean (0.818 vs 0.783) and reduced spread, indicating more consistent segmentation performance.

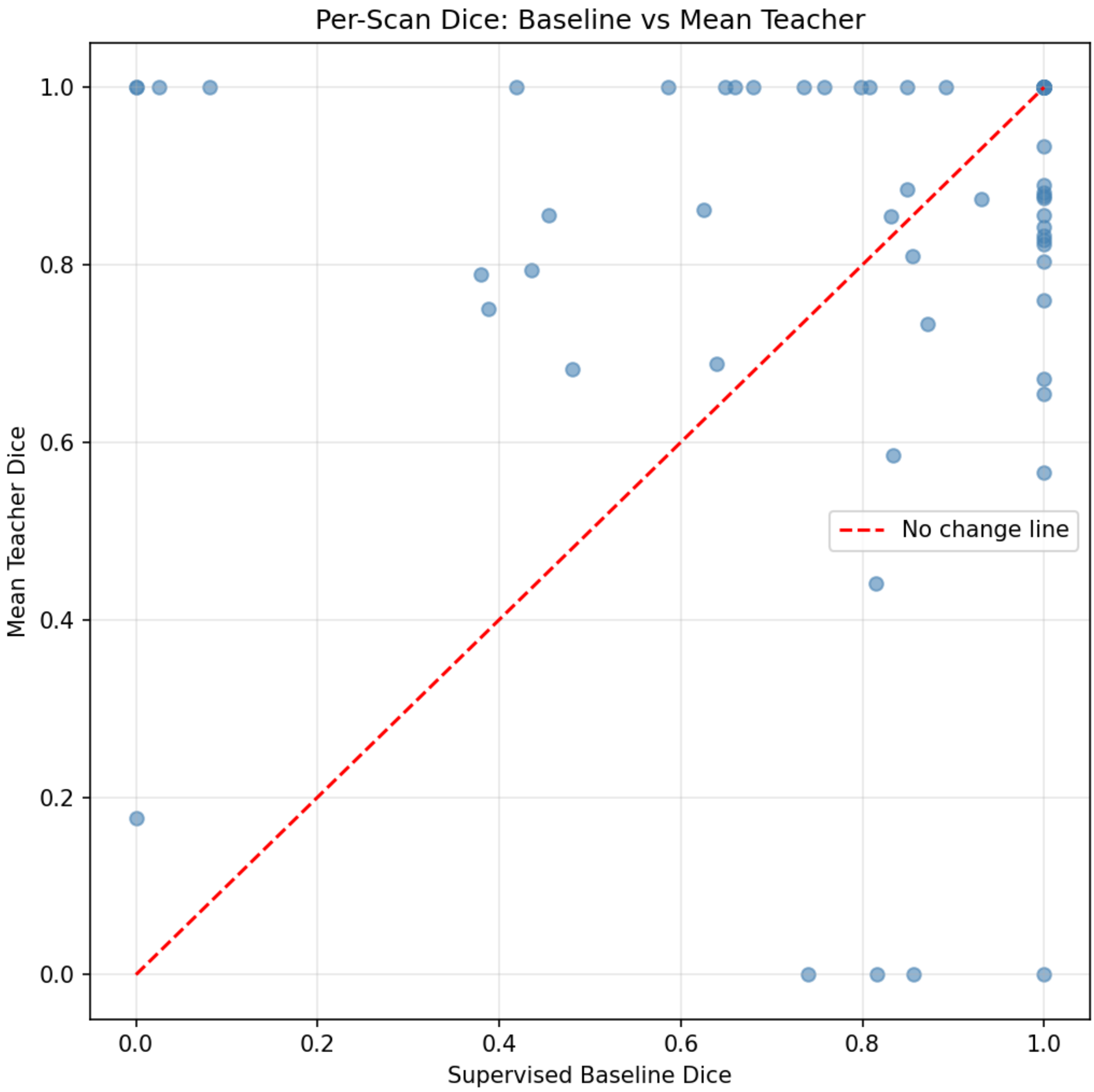
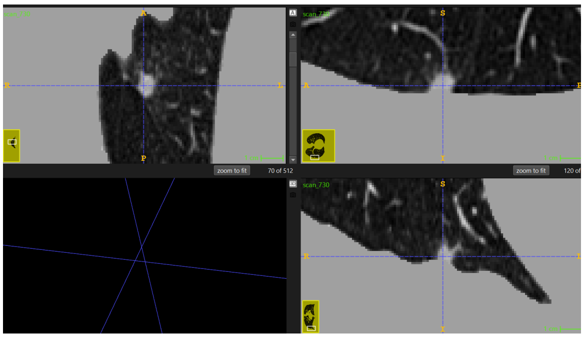
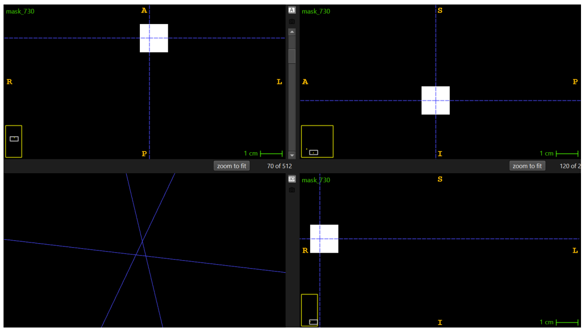


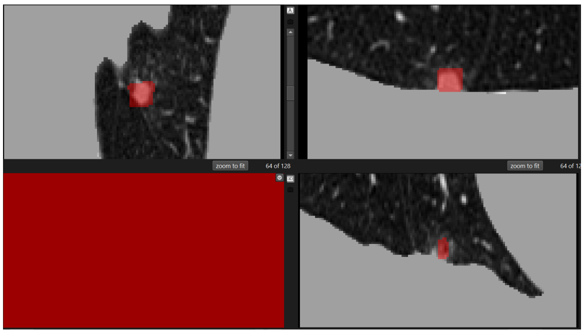
Figure 15: Per-scan Dice: Supervised Baseline (x-axis) vs Mean Teacher (y-axis). The points above the dashed diagonal indicate scans where Mean Teacher outperformed the baseline. We can see the majority of partial-overlap scans favour Mean Teacher.



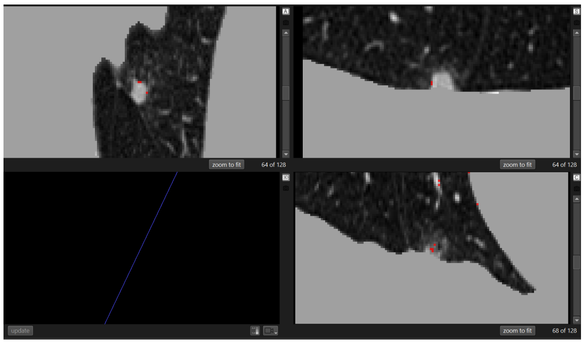
(a) Raw CT scan with ground truth annotation



(b) Ground truth mask from preprocessing



(c) Supervised Baseline (Dice = 0.4521, epoch 100)



(d) Mean Teacher (Dice = 0.6617, best checkpoint)

Figure 16: Qualitative segmentation comparison on validation scan_730. (a) Raw CT scan showing the nodule as a small bright region within the lung parenchyma. (b) Cubic ground truth mask generated during preprocessing where the bounding box shape reflects a known limitation of the annotation pipeline. (c) Supervised Baseline prediction at epoch 100 (Dice 0.4521), showing a larger, less precise segmentation region. (d) Mean Teacher best checkpoint prediction (0.6617), producing a more compact and accurately localised segmentation consistent with the nodule boundary. The qualitative improvement in (d) is consistent with the quantitative ASD reduction of 29.7% observed in Table 6

0.6617, visualised across axial, sagittal and coronal planes in 3D Slicer. From Figure (d) in 16, we can see that the Mean Teacher prediction produces a smaller, visibly more precisely localised segmentation compared to the Baseline in (c), which over-predicts the region with a larger and less focused mask. This qualitative difference is consistent with the quantitative improvements in HD95 ($5.12 \rightarrow 4.05$ voxels) and ASD ($1.55 \rightarrow 1.09$ voxels) reported in Table 6, both of which penalise boundary inaccuracy and over-segmentation.

The ground truth mask in (b) takes a cubic shape rather than an ellipsoidal approximation of the nodule which I mentioned previously in Section 4.2 as a limitation of the preprocessing pipeline. Unfortunately, nodule annotations were generated as bounding cubes from the centre coordinates and diameter provided in annotations.csv. The imprecision of the ground truth boundary factors for the high standard deviation in HD95 across the validation set explains cases where a visually reasonable prediction still scores low Dice due to insufficient overlap with the cubic ground truth region.

Chapter 6

Discussion

6.1 Benefits and Limitations of the Approach

6.1.1 Benefits

The main benefit of the Mean Teacher framework shown here is its ability to leverage unlabelled data to boost segmentation performance, without any additional annotation cost. The 135 unlabelled scans used in training required no manual labelling, yet including them produced measurable improvements across all four metrics. This is especially valuable in the medical imaging, where expert annotation is time-consuming, and expensive.

The EMA-based teacher networkacts like a smoothing filter on the noisy gradients from small-batch 3D training. This smoothing effect help prevents the student network from over-fitting to difficult patches, which is reflected in the lowest standard deviation of Mean Teacher's Dice scores (0.2682 vs 0.2919). The sigmoid ramp-up schedule also helps by letting the student build a solid supervised foundation before the consistency loss kicks in, avoiding corruption from the teacher's early shaky predictions.

Both student and teacher use the same 3D ResUNet architecture, so the Mean Teacher framework adds no extra cost during actual use. At deployment, only the student network runs, keeping clinical workflows fast despite the extra training effort. Setio et al. [2017]

6.1.2 Limitations

Several limitations of the current approach should be acknowledged. First, the evaluation protocol uses the same validation set for both model selection and final evaluation due to a reduced dataset, which may produce optimistic absolute metrics. An evaluation would require a test set that is never used during training. This limitation affects the absolute values reported but not the relative comparison between the two models, since both were evaluated identically.

Second, the ground truth masks were generated as bounding cubes rather than using ellipsoidal to approximate of the nodule shape. Unfortunately as demonstrated in the qualitative analysis, this introduces imprecision into the ground truth boundary, which contributes to the high standard deviation in HD95 (±6.76 for baseline, ±6.37 for Mean Teacher) and causes some visually correct predictions to score low Dice due to geometric mismatch with the cubic ground truth.

Third, the validation metric is computed on randomly sampled patches rather than a full-volume sliding window inference. This allows for significant epoch-to-epoch variance as demonstrated in Figure 12, making it difficult to tell convergence from the training curve alone. Using sliding window inference over the full volume would produce more stable and meaningful validation scores.

Fourth, the dataset used is a subset of LUNA16 (300 labelled, 135 unlabelled scans) rather than the full 888-scan corpus. Whilst this is necessary by the remote compute constraints described in Section 5.1, this does reduce the numerical comparability directly with published benchmarks that use the full dataset.

Fifth, no formal ablation studies were conducted due to the computational cost of each training run (approximately 40-50 hours). Although the mode was still trained, and evaluated, the effect of the EMA smoothing coefficient α, the ramp-up length, and the ratio of labelled to unlabelled data were not separately quantified for this task.

6.2 Further Interpretation of scans

The training curve in Figure 12 reveals several important patterns. Both models converge from near-zero within the first five epochs, which confirms that the biased patch sampling strategy described in Section 4.3.2 was essential. This is shown as without it, both models produced zero Dice for the first 17+ epochs due to random patches rarely containing nodule voxels. The baseline demonstrates high epoch-to-epoch variance throughout training, whilst the Mean Teacher shows progressively more stable from approximately epoch 56 onward. This stabilisation coincides with the sigmoid ramp-up schedule finishing at epoch 20 and the addition of reliable EMA signal in the teacher network. Both are consistent with the expectation that the teacher requires sufficient training before its pseudo-labels become informative.

Both models produced Dice = 0.0 on a small number of scans. There are two likely causes of the fail cases. First, the random patch sampling strategy occasionally samples a patch that does not contain the nodule, which results in an empty prediction that scores zero regardless of model quality. Second, the cubic ground truth masks generated during preprocessing may not overlap correctly with an ellipsoidal model prediction even when the prediction is spatially correct, which produces a low Dice score. These failure cases are therefore due to evaluation methodology rather than model failure.

6.3 Comparison with Related Work

Several LUNA16 studies use overlap and boundary metrics such as Dice, Jaccard, Hausdorff distance, and ASD, but not all papers use the same full set of metrics. For fair comparison, this work focuses on papers that report matching segmentation metrics on LUNA16, including MRUNet-3D and related LUNA16 nodule segmentation studies.

Table 7: Comparison of nodule segmentation performance on LUNA16. Published results use the full 888-scan dataset with cross-validation. My results use 300 labelled scans with an 80/20 validation split. ASD converted from voxels to mm using 0.7mm voxel spacing. ↑ higher is better, ↓ lower is better.

Model	**Supervision**	**DSC (%) ↑**	**ASD (mm) ↓**
V-Net Milletari et al. [2016]	Full	78.14 ± 13.51	0.55 ± 1.22
3D UNet Çiçek et al. [2016]	Full	81.95 ± 12.07	0.43 ± 0.96
UNet++ Zhou et al. [2018]	Full	82.42 ± 10.93	0.38 ± 0.74
MRUNet-3D Bbosa et al. [2024]	Full	**83.47 ± 10.64**	**0.35 ± 0.87**
MY Supervised Baseline	Full	78.27 ± 29.19	1.09
My Mean Teacher (SSL)	Semi	81.77 ± 26.82	0.76

Notably, the Mean Teacher framework achieves a DSC of 81.77%, comparable to fully supervised 3D UNet (81.95%) and UNet++ (82.42%) trained on the complete LUNA16 dataset Bbosa, Gui, Luo, Liu, Efio-Akolly, and Chen [2024], despite using only 300 labelled scans. This suggests that consistency regularisation on unlabelled data can partially compensate for reduced label availability, supporting the core hypothesis of this work. More recent multi-stage frameworks combining segmentation with detection and classification, such as the multi-phase Swin Transformer approach of Dhayalini and Revathi Alias Ponmozhi [2026], report substantially higher Dice scores of 97–98% on LUNA16 using powerful CNN-Transformer backbones. However, these are not directly comparable to our work, which focuses on a plain 3D ResUNet under semi-supervised constraints rather than architectural enhancement.

Once again, it should be noted that my evaluation used the same validation set for both model selection and final evaluation, which may produce optimistic absolute metrics, but give a reasonable prediction. A rigorous comparison with published results would mean after training is complete we have a fully held-out test set evaluated on top.

6.4 Implications for Clinical Practice

The results of this work have several implications for the clinical deployment of automated lung nodule segmentation systems. The most important takeaway is that strong segmentation performance can still be achieved with a relatively small labelled dataset by making use of unlabelled scans. In clinical practice, expert annotation of CT scans takes a lot of time and effort, and this is often the main barrier to deploying deep learning systems Shak et al. [2020]. The Mean Teacher framework provides a practical way to ease this problem by learning from the large amount of unlabelled CT data that hospitals already collect during routine screening.

The 29.7% reduction in ASD achieved by Mean Teacher is particularly important clinically. Accurate nodule boundary delineation is important for nodule volume measurement, as they guide Lung-RADS and Fleischner Society guidelines for determining follow-up intervals and when to intervine Siegel et al. [2022]. A model that more precisely draws the nodule boundary edges reduces the risk of classifying a nodule in the wrong size category, which could otherwise lead to unnecessary scans or, worse, missed malignancies.

However, there are several limitations that woudl need to be addressed before this approach could be considered for clinical deployment. The high standard deviation in Dice (± 0.27 for Mean Teacher) indicates inconsistent performance across individual scans, which is unacceptable as in a clinical context reliability on every single case matters just as much as good average results. The cubic ground truth masks used as ground truth would need to be replaced with radiologist-drawn contours or ellipsoids to produce trustworthy ground training data. Finally, the model has not been tested on an independent data set or across different scanner types, radiation dose levels, or patient demographics, which are all essential steps before it could be approved for real clinical use.

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This project developed and evaluated a semi-supervised medical image segmentation pipeline for lung nodule detection using the LUNA16 dataset. The key contributions are as follows:

1. A supervised 3D ResUNet baseline trained on 300 labelled CT scans, achieving a mean Dice of 0.7827 on the validation set.
2. A Mean Teacher semi-supervised framework trained on the same 300 labelled scans supplemented by 135 unlabelled scans, achieving a mean Dice of 0.8177, a 3.5% improvement over the baseline.
3. A reproducible preprocessing, training, and evaluation pipeline including foreground-biased patch sampling, HU normalisation, EMA-based teacher updates, and sigmoid ramp-up consistency weighting.
4. A comprehensive four-metric evaluation (Dice, Jaccard, HD95, ASD) with qualitative visualisation in 3D Slicer, demonstrating that Mean Teacher produces more precise nodule boundary delineation alongside improved volumetric overlap.

7.2 Overall remarks

This work demonstrates that the Mean Teacher consistency regularisation framework can meaningfully improve lung nodule segmentation performance over a supervised baseline when unlabelled data is available. Across all four evaluation metrics, Mean Teacher outperformed the supervised baseline, with the most clinically significant improvement being a 29.7% reduction in Average Surface Distance (1.55 $\rightarrow$ 1.09 voxels), indicating substantially better boundary quality.

The central finding is that Mean Teacher achieved a Dice of 81.77%, comparable to fully supervised 3D UNet (81.95%) and UNet++ (82.42%) from the literature Bbosa et al. [2024], despite using only 300 labelled scans versus the full 888-scan datasets used in those studies. This supports the core hypothesis: that consistency regularisation on unlabelled data can partially compensate for reduced label availability, making semi-supervised learning a practical approach in data-scarce medical imaging settings.

Several limitations were identified. The evaluation used the same validation set for model selection and final evaluation, which may produce optimistic absolute metrics. The ground truth masks are cubic bounding boxes rather than ellipsoidal nodule approximations, introducing imprecision that contributes to high HD95 variance. The dataset used is a subset of LUNA16 due to compute constraints, and no formal ablation studies were conducted.

Despite these limitations, the results provide a valid and reproducible demonstration that Mean Teacher semi-supervised learning is an effective strategy for lung nodule segmentation when labelled data is limited.

7.3 Future Improvements

The cubic bounding box masks could be improved by using the original CT spacing metadata to generate proper ellipsoidal masks. This would fix the high HD95 variance we saw and give much cleaner ground truth. I'd just need to run the fix_masks.py script across the whole dataset.

Random patch sampling during validation gives inconsistent results between epochs. Switching to full-volume sliding window inference would produce stable, clinically realistic evaluation scores like you'd see in real papers.

There's also the uncertainty-aware Mean Teacher from Yu et al. [2019b] which only uses high-confidence teacher predictions. This consistently beats the vanilla version in other studies, so that's an obvious upgrade path.

Proper ablation studies would be valuable too - testing different EMA α values, ramp-up lengths and label ratios (10%, 20%, 50%). You'd really see where the unlabelled data adds the most value.

Training on the full 888-scan LUNA16 with proper train/val/test splits would let us match literature numbers directly. Adding standard augmentations like flips and rotations would help both models equally.

Finally, before anything clinical, you'd want testing across different scanner manufacturers, dose levels, patient demographics, and proper radiologist contours rather than algorithm-generated bounding boxes.

References

D. R. Aberle, C. D. Berg, W. C. Black, and et al. The national lung screening trial: overview and study design. *Radiology*, 258:243–253, 2011. doi: 10.1148/radiol.10091808.

M. Al-Shabi, B. L. Lan, W. Y. Chan, K. H. Ng, and M. Tan. Lung nodule classification using deep local–global networks. *International Journal of Computer Assisted Radiology and Surgery*, 14(10):1815–1819, 2019a. doi: 10.1007/s11548-019-01981-7.

Mundher Al-Shabi, Hong Kyu Lee, and Maxine Tan. Gated-dilated networks for lung nodule classification in ct scans. *IEEE Access*, 7:178827–178838, 2019b. doi: 10.1109/ACCESS.2019.2958663.

Mundher Al-Shabi, Kelvin Shak, and Maxine Tan. Procan: Progressive growing channel attentive non-local network for lung nodule classification. *Pattern Recognition*, 122:108309, 2022. doi: 10.1016/j.patcog.2021.108309.

Richard H. Albert and Jill J. Russell. Evaluation of the solitary pulmonary nodule. *American Family Physician*, 80(8):827–831, 2009.

Ignacio Arganda-Carreras, Srinivas C. Turaga, Daniel R. Berger, Dan Cireşan, Alessandro Giusti, Luca M. Gambardella, Jürgen Schmidhuber, Dmitry Laptev, Sarvesh Dwivedi, Joachim M. Buhmann, Ting Liu, Mojtaba Seyedhosseini, Tolga Tasdizen, Lee Kamentsky, Radim Burget, Vaclav Uher, Xiao Tan, Changming Sun, Tuan D. Pham, Erhan Bas, Mustafa G. Uzunbas, Albert Cardona, Johannes Schindelin, and H. Sebastian Seung. Crowdsourcing the creation of image segmentation algorithms for connectomics, 2015. ISSN 1662-5129. URL `https://www.frontiersin.org/journals/neuroanatomy/articles/10.3389/fnana.2015.00142`.

Samuel G. Armato, Geoffrey McLennan, Luc Bidaut, and et al. The lung image database consortium (lidc) and image database resource initiative (idri): A completed reference database of lung nodules on ct scans. *Medical Physics*, 38:915–931, 2011. doi: 10.1118/1.3528204.

Philip Bachman, Ouais Alsharif, and Doina Precup. Learning with pseudo-ensembles. 2014. URL `https://arxiv.org/abs/1412.4864`.

Hritam Basak and Zhaozheng Yin. Pseudo-label guided contrastive learning for semi-supervised medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19786–19796, 2023.

Ronald Bbosa, Hao Gui, Fei Luo, Feng Liu, Kafui Efio-Akolly, and Yi-Ping Phoebe Chen. Mrunet-3d: A multi-stride residual 3d unet for lung nodule segmentation. *Methods*, 226:89–101, 2024. ISSN 1046-2023. doi: https://doi.org/10.1016/j.ymeth.2024.04.008. URL `https://www.sciencedirect.com/science/article/pii/S1046202324000926`.

David Berthelot, Nicholas Carlini, Ekin D. Cubuk, Alexey Kurakin, Kihyuk Sohn, Han Zhang, and Colin Raffel. Re-mixmatch: Semi-supervised learning with distribution alignment and augmentation anchoring. *arXiv preprint arXiv:1911.09785*, 2019.

Cell Tracking Challenge. Cell tracking challenge, 2014. URL `http://celltrackingchallenge.net/`. Accessed: 2026-04-06.

Jun Chen, Heye Zhang, Raad Mohiaddin, Tom Wong, David Firmin, Jennifer Keegan, and Guang Yang. Adaptive hierarchical dual consistency for semi-supervised left atrium segmentation on cross-domain data. *IEEE Transactions on Medical Imaging*, 41(2): 420–433, 2022. doi: 10.1109/TMI.2021.3113678.

Bhupender S. Chhikara and Keykavous Parang. Global cancer statistics 2022: The trends projection analysis. *Chemical Biology Letters*, 10(1):451, 2023. URL `https://pubs.thesciencein.org/journal/index.php/cbl/article/view/451`.

Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, and Olaf Ronneberger. 3d u-net: Learning dense volumetric segmentation from sparse annotation. In Sebastien Ourselin, Leo Joskowicz, Mert R. Sabuncu, Gozde Unal, and William Wells, editors, *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2016*, pages 424–432, Cham, 2016. Springer International Publishing. ISBN 978-3-319-46723-8.

Dan C. Ciresan, Luca M. Gambardella, Alessandro Giusti, and Jürgen Schmidhuber. Deep neural networks segment neuronal membranes in electron microscopy images. In *Advances in Neural Information Processing Systems*, pages 2852–2860, 2012.

codegenes. A comprehensive guide to 3d u-net in pytorch, 2024. URL `https://codegenes.net/3d-u-net-pytorch-guide/`. Accessed: 2026-04-19.

Wenbo Cui, Yonghuai Liu, Yuanyuan Li, Mingliang Guo, Yan Li, Xiaoxiao Li, Tian Wang, Xiaoyang Zeng, and Chuanxin Ye. Semi-supervised brain lesion segmentation with an adapted mean teacher model. In *International Conference on Information Processing in Medical Imaging*, IPMI, pages 554–565, 2019.

M. Dhayalini and B. Revathi Alias Ponmozhi. Multi-phase deep learning framework with multiscale adaptive swin transformer and embedding attention for precision lung nodule detection and classification. *Scientific Reports*, 16:1652, January 2026. doi: 10.1038/s41598-025-31147-2. URL `https://doi.org/10.1038/s41598-025-31147-2`.

Alexey Dosovitskiy, Jost Tobias Springenberg, Martin Riedmiller, and Thomas Brox. Discriminative unsupervised feature learning with convolutional neural networks. In *Advances in Neural Information Processing Systems*, 2014.

Yu Duan, Junzhi He, Runxin Zhang, Rong Wang, Xuelong Li, and Feiping Nie. Prediction consistency regularization for generalized category discovery. *Information Fusion*, 112: 102547, 2024. ISSN 1566-2535. doi: https://doi.org/10.1016/j.inffus.2024.102547. URL `https://www.sciencedirect.com/science/article/pii/S1566253524003257`.

Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 1050–1059, 2016.

Guohong Han, Xiaobo Liu, Guangyao Zheng, Minghui Wang, and Sha Huang. Automatic recognition of 3d ggo ct imaging signs through the fusion of hybrid resampling and layer-wise fine-tuning cnns. *Medical & Biological Engineering & Computing*, 56(12): 2201–2212, 2018. doi: 10.1007/s11517-018-1850-z.

Guohong Han, Xiaobo Liu, Hao Zhang, Guangyao Zheng, Noor Qamar Soomro, Minghui Wang, and Wei Liu. Hybrid resampling and multi-feature fusion for automatic recognition of cavity imaging sign in lung ct. *Future Generation Computer Systems*, 99: 558–570, 2019. doi: 10.1016/j.future.2019.05.009.

Guotai He. Lung ct imaging sign classification through deep learning on small data. *arXiv preprint arXiv:1903.00183*, 2019. URL `https://arxiv.org/abs/1903.00183`.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015. URL `https://arxiv.org/abs/1512.03385`.

Yangqing Jia, Evan Shelhamer, Jeff Donahue, Sergey Karayev, Jonathan Long, Ross Girshick, Sergio Guadarrama, and Trevor Darrell. Caffe: Convolutional architecture for fast feature embedding. *Proceedings of the ACM International Conference on Multimedia*, 2014.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization, 2017. URL `https://arxiv.org/abs/1412.6980`.

Samuli Laine and Timo Aila. Temporal ensembling for semi-supervised learning. 2017. URL `https://arxiv.org/abs/1610.02242`.

Shumeng Li, Ziyuan Zhao, Kaixin Xu, Zeng Zeng, and Cuntai Guan. Hierarchical consistency regularized mean teacher for semi-supervised 3d left atrium segmentation. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, page 3395–3398. IEEE, November 2021a. doi: 10.1109/embc46164.2021.9629941. URL `http://dx.doi.org/10.1109/EMBC46164.2021.9629941`.

Xiaomeng Li, Lequan Yu, Hao Chen, Chi-Wing Fu, Lei Xing, and Pheng-Ann Heng. Transformation-consistent self-ensembling model for semisupervised medical image segmentation. *IEEE Transactions on Neural Networks and Learning Systems*, 32(2):523–534, 2021b. doi: 10.1109/TNNLS.2020.2995319.

Xiaomeng Li, Lequan Yu, Hao Chen, Chi-Wing Fu, Lei Xing, and Pheng-Ann Heng. Transformation-consistent self-ensembling model for semisupervised medical image segmentation. *IEEE Transactions on Neural Networks and Learning Systems*, 32(2):523–534, 2021c. doi: 10.1109/TNNLS.2020.2995319.

Yutong Li, Yan Li, and Yong Fan. Deepseed: 3d squeeze-and-excitation encoder-decoder convolutional neural networks for pulmonary nodule detection. In *2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI)*, pages 1866–1869. IEEE, 2020. doi: 10.1109/ISBI45749.2020.9098503.

Ming Liu, Xu Liu, Jian Jiang, Yu Liu, Fan Fan, Hong Zhao, and Aiping Zhou. A semi-supervised convolutional transfer neural network for 3d pulmonary nodule detection. *Neurocomputing*, 391:199–209, 2019. doi: 10.1016/j.neucom.2018.12.081.

Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3431–3440, 2015.

Xiangde Luo, Jieneng Chen, Tao Song, and Guotai Wang. Semi-supervised medical image segmentation through dual-task consistency. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(10):8801–8809, May 2021. doi: 10.1609/aaai.v35i10.17066. URL `https://ojs.aaai.org/index.php/AAAI/article/view/17066`.

Ling Ma, Xiabi Liu, Li Song, Chunwu Zhou, Xiaoyan Zhao, and Yong Zhao. A new classifier fusion method based on historical and on-line classification reliability for recognizing common ct imaging signs of lung diseases. *Computerized Medical Imaging and Graphics*, 40:39–48, 2015. doi: 10.1016/j.compmedimag.2014.12.004.

Ling Ma, Xiabi Liu, Yong Gao, Yong Zhao, Xiaoyan Zhao, and Chunwu Zhou. A new method of content-based medical image retrieval and its applications to ct imaging sign retrieval. *Journal of Biomedical Informatics*, 66:148–158, 2017. doi: 10.1016/j.jbi.2017.01.002.

Ling Ma, Xiabi Liu, and Baowei Fei. A multi-level similarity measure for the retrieval of the common ct imaging signs of lung diseases. *Medical & Biological Engineering & Computing*, 58(5):1015–1029, 2020. doi: 10.1007/s11517-020-02139-6.

Martin Maška, Vladimír Ulman, David Svoboda, Petr Matula, Pavel Matula, Cristina Ederra, Ainhoa Urbiola, Tino España, Subramanian Venkatesan, Dana Balak, et al. A benchmark for comparison of cell tracking algorithms. *Bioinformatics*, 30(11):1609–1617, 2014. doi: 10.1093/bioinformatics/btu080.

Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *2016 Fourth International Conference on 3D Vision (3DV)*, pages 565–571, 2016. doi: 10.1109/3DV.2016.79.

Takeru Miyato, Shin-ichi Maeda, Masanori Koyama, and Shin Ishii. Virtual adversarial training: A regularization method for supervised and semi-supervised learning. *arXiv preprint arXiv:1704.03976*, 2017.

Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks, 2018. URL `https://arxiv.org/abs/1802.05957`.

Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992. ISSN 0363-0129. doi: 10.1137/0330046.

Shanshan Qiu, Jie Sun, Ting Zhou, Guoxiang Gao, Zhen He, and Tao Liang. Spiculation sign recognition in a pulmonary nodule based on spiking neural p systems. *BioMed Research International*, 2020:6619076, 2020. doi: 10.1155/2020/6619076.

Vishal Rajput. Robustness of different loss functions and their impact on networks learning capability, 2021. URL `https://arxiv.org/abs/2110.08322`.

Antti Rasmus, Mathias Berglund, Mikko Honkala, Harri Valpola, and Tapani Raiko. Semi-supervised learning with ladder networks. In *Advances in Neural Information Processing Systems*, volume 28, pages 3546–3554. Curran Associates, Inc., 2015.

Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. *arXiv preprint arXiv:1506.02640*, 2015. URL `http://arxiv.org/abs/1506.02640`.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi, editors, *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241. Springer, 2015. doi: 10.1007/978-3-319-24574-4_28.

Mehdi Sajjadi, Mehran Javanmardi, and Tolga Tasdizen. Regularization with stochastic transformations and perturbations for deep semi-supervised learning. In *Advances in Neural Information Processing Systems*, volume 29, pages 1163–1171. Curran Associates, Inc., 2016.

Arnaud A. A. Setio, Alberto Traverso, Thomas de Bel, and et al. Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The luna16 challenge. *Medical Image Analysis*, 2017. doi: 10.1016/j.media.2017.08.021.

Kelvin Shak, Mundher Al-Shabi, Andrea Liew, Boon Leong Lan, Wai Yee Chan, Kwan Hoong Ng, and Maxine Tan. A new semi-supervised self-training method for lung cancer prediction. 2020. URL `https://arxiv.org/abs/2012.09472`.

Rebecca L. Siegel, Kimberly D. Miller, Hannah E. Fuchs, and Ahmedin Jemal. Cancer statistics, 2022. *CA: A Cancer Journal for Clinicians*, 72(1):7–33, 2022. doi: 10.3322/caac.21708. URL `https://acsjournals.onlinelibrary.wiley.com/doi/abs/10.3322/caac.21708`.

Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014. ISSN 1532-4435.

Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. 2018. URL `https://arxiv.org/abs/1703.01780`.

Bob D Vos et al. Deep learning in pulmonary nodule detection and segmentation: a systematic review. *European Radiology*, 2024. doi: 10.1007/s00330-024-10907-0.

Ping Wang, Jizong Peng, Marco Pedersoli, Yuanfeng Zhou, Caiming Zhang, and Christian Desrosiers. Self-paced and self-consistent co-training for semi-supervised image segmentation. *Medical Image Analysis*, 73:102146, 2021a. ISSN 1361-8415. doi: https://doi.org/10.1016/j.media.2021.102146. URL `https://www.sciencedirect.com/science/article/pii/S1361841521001924`.

Yuyang Wang, Peng Tang, Yong Zhou, Wenqi Shen, Elliot K. Fishman, and Alan L. Yuille. Learning inductive attention guidance for partially supervised pancreatic ductal adenocarcinoma prediction. *IEEE Transactions on Medical Imaging*, 40(10):2723–2735, 2021b.

Botong Wu, Zhen Zhou, Jian Wang, and Yizhou Wang. Joint learning for pulmonary nodule segmentation, attributes and malignancy prediction. In *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI)*, pages 1109–1113, Washington, DC, USA, April 2018. IEEE. doi: 10.1109/ISBI.2018.8363764.

Qizhe Xie, Zihang Dai, Eduard Hovy, Minh-Thang Luong, and Quoc V. Le. Unsupervised data augmentation for consistency training. In *Advances in Neural Information Processing Systems*, volume 33, pages 6256–6268, 2020a.

Qizhe Xie, Minh-Thang Luong, Eduard Hovy, and Quoc V. Le. Self-training with noisy student improves imagenet classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2020b. doi: 10.1109/CVPR42600.2020.01070.

Yue Xie, Yong Xia, Jian Zhang, Yubo Song, Dan Feng, Michael Fulham, and Weidong Cai. Knowledge-based collaborative deep learning for benign-malignant lung nodule classification on chest ct. *IEEE Transactions on Medical Imaging*, 38(4):991–1004, 2019. doi: 10.1109/TMI.2018.2876510.

Xiangli Yang, Zixing Song, Irwin King, and Zenglin Xu. A survey on deep semi-supervised learning. *IEEE Transactions on Knowledge and Data Engineering*, 35(9):8934–8954, 2023. doi: 10.1109/TKDE.2022.3220219.

Xavier Yorke. MISNet: Implementation of Residual U-Net in Python. `https://github.com/XavierYorke/MISNet`, 2024.

Lequan Yu, Shujun Wang, Xiaomeng Li, Chi-Wing Fu, and Pheng-Ann Heng. Uncertainty-aware self-ensembling model for semi-supervised 3d left atrium segmentation. In Dinggang Shen, Tianming Liu, Terry M. Peters, Lawrence H. Staib, Caroline Essert, Sean Zhou, Pew-Thian Yap, and Ali Khan, editors, *Medical Image Computing and Computer Assisted Intervention – MICCAI 2019*, pages 605–613, Cham, 2019a. Springer International Publishing. ISBN 978-3-030-32245-8.

Lequan Yu, Shun Wang, Xiaomeng Li, Chi-Wing Fu, and Pheng-Ann Heng. Uncertainty-aware self-ensembling model for semi-supervised 3d left atrium segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, MICCAI, pages 605–613, 2019b.

Ketian Zhan, Yunpeng Wang, Yaoyao Zhuo, Yi Zhan, Qinqin Yan, Fei Shan, Lingxiao Zhou, Xinrong Chen, and Lei Liu. An uncertainty-aware self-training framework with consistency regularization for the multilabel classification of common computed tomography signs in lung nodules. *Quantitative Imaging in Medicine and Surgery*, 13(9), 2023. ISSN 2223-4306. URL `https://qims.amegroups.org/article/view/115262`.

Jian Zhang, Yue Zhang, Yuxiang Xia, and Hui Zeng. Nodule: Combining constrained multi-scale log filters with densely dilated 3d deep convolutional neural network for pulmonary nodule detection. *Neurocomputing*, 317:159–167, 2018. doi: 10.1016/j.neucom.2018.08.022.

Yichi Zhang and Jianpeng Zhang. Dual-task mutual learning for semi-supervised medical image segmentation. In *Pattern Recognition and Computer Vision*, volume 13021 of *PRCV*, pages 548–559, 2021.

Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: A nested u-net architecture for medical image segmentation. In Danail Stoyanov, Zeike Taylor, Gustavo Carneiro, Tanveer Syeda-Mahmood, Anne Martel, Lena Maier-Hein, João Manuel R.S. Tavares, Andrew Bradley, João Paulo Papa, Vasileios Belagiannis, Jacinto C. Nascimento, Zhi Lu, Sailesh Conjeti, Mehdi Moradi, Hayit Greenspan, and Anant Madabhushi, editors, *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, pages 3–11, Cham, 2018. Springer International Publishing. ISBN 978-3-030-00889-5.

Wentao Zhu, Chaochun Liu, Wei Fan, and Xiaohui Xie. Deeplung: Deep 3d dual path nets for automated pulmonary nodule detection and classification. In *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 673–681, 2018. doi: 10.1109/WACV.2018.00079.